A REPORT ON

STRUCTURAL ANALYSIS & DESIGN

OF

RESIDENTIAL BUILDING

FOR

MR. NHUCHHE RAM LAWAJU

Location: Bhaktapur Municipality

Aug, 2026

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DECLERATION OF REPORT

I, consultant/designer issuing this repudiation letter on behalf of final structural analysis and design report for **MR. Nhuchhe Ram Lawaju** and after construction of proposed super structure.

The site supervisions and quality test of the different constructions materials of the proposed building do not undertaken by the consultant. Hence any failure or error occurred with the building due to non- implementation of the recommendations made in this report and structural drawing or due to improper construction will not be the responsibility of the consultant/designer.

The structural analysis has been completed on the assumptions that the proposed building shall be constructed under the proper supervisions of skilled technical manpower.
The designer will not be responsible if any alterations to the structural system is made by the client or the contractor without the prior written permission from the designer, or the alterations to the non-structural system is made such that the weight of each individual floor or the weight of the building.

Designed by

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1 INTRODUCTION

1.1 AN OVERVIEW OF EARTHQUAKE-RESISTANT DESIGN

Nepal lies in an earthquake prone zone. It is located in the boundary between Indian plate and Tibetan plate. Thus, Nepal experiences earthquakes frequently. Earthquakes are the most sudden, dramatic and devastating natural calamities. Although they last only for few seconds to minutes, they are the most devastating ones. Past records of the earthquake show that eighteen major earthquakes have hit Nepal till now. So, special consideration of earthquake forces while designing a building is mandatory in Nepal.

Designing building against earthquake forces does not mean that we are making the building earthquake-proof. Although earthquake proof structures may be viable in terms of engineering but they are not financially viable. However, it is possible to design and build earthquake-resilient (more precisely seismic resilient) structures. According to this philosophy, no matter how much severe earthquake occurs, the building won't collapse although it may be irreparably damaged. In, this background, it is ensured that this building has been designed to withstand all the acting loads including the earthquake forces making it seismic resistant.

The structural design alone is not enough to ensure the safety of a building, equally important, is its construction. The role of a contractor is of paramount importance as he is the one to execute the construction work at site. He is required to execute the work according to drawings supplied by the consultant to him and detailing has to be carefully followed. A large percentage of failure of the building is attributed to poor quality of construction. Past experiences from damages have shown that quality of material and workmanship play an important role in good seismic behavior of the buildings. Hence, quality assurance of construction works is the key to good seismic performance of a building.

Severity of ground shaking at a given location during an earthquake can be minor, moderate and strong. Relatively, minor shakings occur frequently, moderate shakings occasionally and strong shakings rarely. For instance, on average annually about 800 earthquakes of magnitude 5.0-5.9 on Richter scale occur in the world while the number is only about 19 for magnitude of range 7.0-7.9 on Richter scale. So, should we design and construct a building to resist that rare earthquake shaking that may come only once in 500 years or even once in 2000 years at the chosen project site, even though the life of the building itself may be only 50 or 100 years? Since it costs money to provide additional earthquake safety in buildings, a conflict arises: Should we do away with the design of buildings for earthquake effects? Or should we design the buildings to be "earthquake proof" wherein there is no damage during the strong but rare earthquake shaking? Clearly, the former approach can lead to a major disaster, and the second approach is too expensive. Hence, the design philosophy should lie somewhere in between these two extremes.

The engineers do not attempt to make earthquake proof buildings that will not get damaged even during the rare but strong earthquakes; such buildings will be too robust and also too expensive. Instead, the engineering intention is to make buildings earthquake resistant and affordable; such buildings resist the effects of ground shaking, although they may get damaged severely but would not collapse during the strong earthquake. Thus, safety of people and contents is assured in earthquake-resistant buildings, and thereby a disaster is avoided. This is a major objective of seismic design codes throughout the world.

1.2 EARTHQUAKE RESISTANT DESIGN PHILOSOPHY

The earthquake design philosophy may be summarized as follows:

a) Under minor shakings which occur frequently, the main members of the buildings that carry vertical and lateral forces should not be damaged; however non-structural parts may sustain repairable damages. It is assured that the building will be fully operational within a short time and the repair costs will be small
b) Under moderate shakings which are occasional, the structural elements may sustain repairable damage, while the non-structural elements may be heavily damaged; the building will be operational followed by repair and re-strengthening of the damaged structural elements and replacing of damaged non-structural elements.
c) Under strong shakings which are rare, the main members may sustain severe damage, but the building should not collapse; the building may become dysfunctional for further use, but will stand so that people can be evacuated and property recovered.

The degree to which the building is to be assured against the possibility of damage must be considered while designing the building. Important buildings like hospitals, health posts, fire stations, post offices communication towers, electric power houses which play a critical role in post-earthquake activities must remain functional immediately after the earthquake and hence require higher degree of assurance against possible damage. Public buildings, educational buildings, busy commercial complex etc. where the risk of human loss is high and some critical structures like nuclear power plants, dams, etc. whose collapse may cause secondary disasters are also designed to assure higher level of safety against earthquakes. Other buildings like residential buildings, offices, mercantile buildings etc. are designed for lower level of safety, while the design philosophy mentioned above is always maintained. However, the level of safety to be assured in the design may also depend upon the demand of the client.

Design of buildings to resist earthquakes involves controlling the damage to acceptable levels at a reasonable cost. Contrary to the common thinking that any crack in the building after an earthquake means the building is unsafe for habitation, some damage in non-structural elements may be unavoidable. Different types of damages (mainly visualized through cracks; especially so in concrete and masonry buildings) may occur in buildings during earthquakes. Some of these cracks are acceptable (in terms of both their size and location), while others are not. In general, qualified technical professionals may only be knowledgeable of the causes and severity of damage in earthquake-resistant buildings..

2 DESCRIPTION OF THE BUILDING

2.1 BUILDING DESCRIPTION

The building to be analyzed and designed here is a Five storeyResidential building proposed to be constructed for Mr. Nhuchhe Ram Lawaju. The proposed site is located at Bhaktapur Municipality. The Typical floor plan of the building is presented below.

The building description is presented in table below:

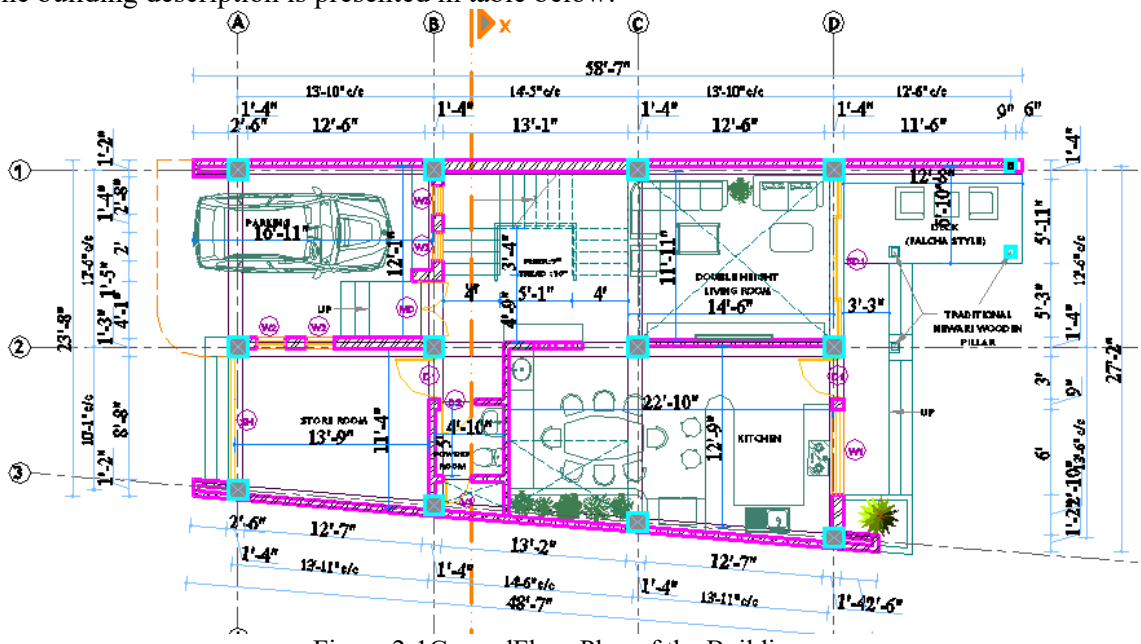


Figure 2-1GroundFloor Plan of the Building

Table 2-1: Building Description

General features	
Building Type	Residential Building
Location	Bhaktapur Municipality
Ground Floor Area	1214.34 . ft.
Architectural features	
Number of storey	Five Storey
Floor Height	9'0" for all floors
Total Height of the Building:	45'0" up to top level from plinth level
Building Lateral Dimensions (Maximum)	Along X: 48'-7"
	Along Y: 27'-2"
Wall and Partition	Brick Masonry in Concrete Mortar outside walls as well as partition walls
Structural features	
Structural System	Special Moment Resisting RCC Frame Structure
Foundation Type	Isolated Footing, Strap Beam & Combine footing
Columns	Rectangular RCC Sections
Beams	Rectangular RCC Sections
Slab	RCC Sections
Geotechnical Features	

Soil Test	Available
Soil Type (assumed)	Type II (Soft Soil)
Seismic Zone considered:	D (as per NBC 105:2025)
Allowable bearing capacity assumed	120 KN/m^2
Material	
Grade of concrete:	M20 for column &beam, M20 for slab, staircase, intermediate beam, fascia beam and footings.
Grade of steel:	Fe 500
Unit weight of concrete	20 KN/m^3
Unit weight of the masonry wall	19.2 KN/ m^3
Unit weight of cement sand plaster	20.4 KN/ m^3
Young's Modulus of Elasticity	5000 $\sqrt{f_{ck}}$
Poisson's Ratio	0.20 for concrete
	0.30 for rebar

3 NUMERICAL MODELING

A three-dimensional finite element model of the whole structural system is created in SAP 2000. The structural system is considered as a Special Moment Resisting Frame structure. Main components to be modeled are: Beams, Columns & Slabs. Beams and columns are modeled as frame elements while slabs are modeled as four-nodded shell elements. The support condition at the base is idealized as a fixed support and hence any moment release at the base of columns is neglected. The moment release at the beam/column joints is also neglected.

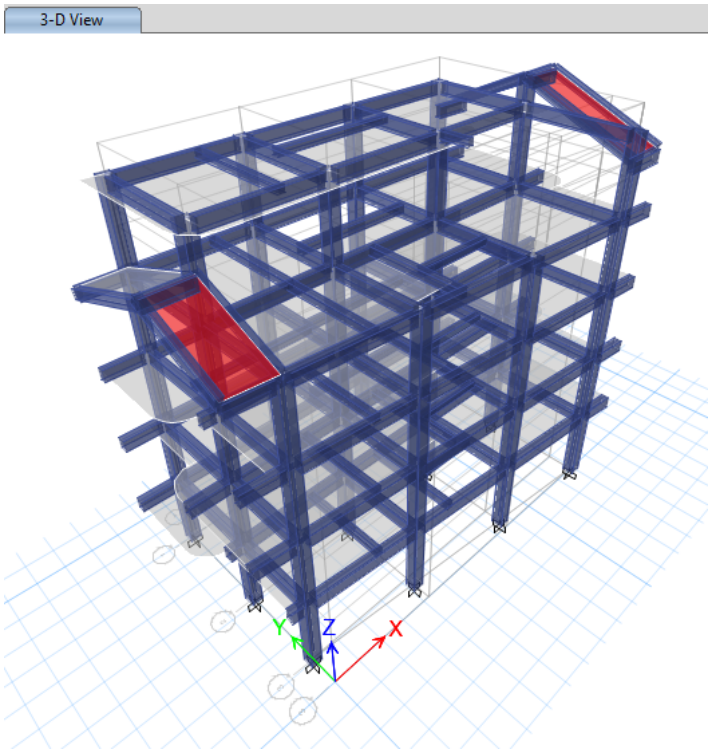


Figure 3-1: Three-Dimensional view of Model created in Etabs

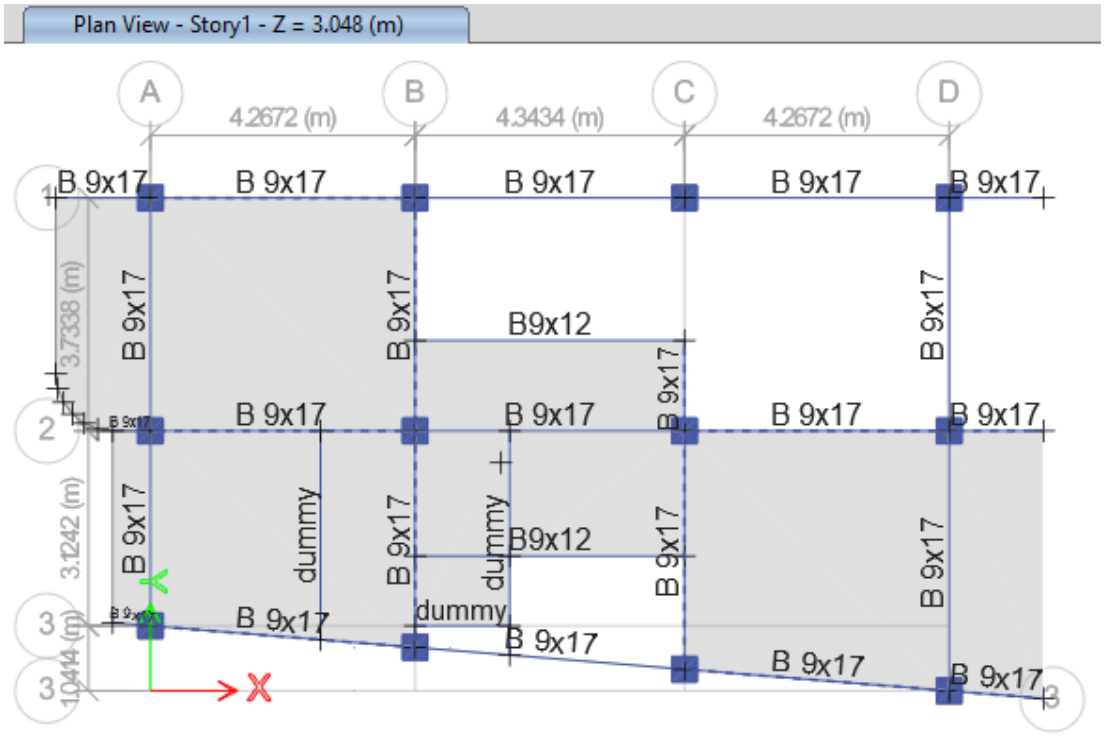


Figure 3-2: First Floor Plan of Model created in Etabs

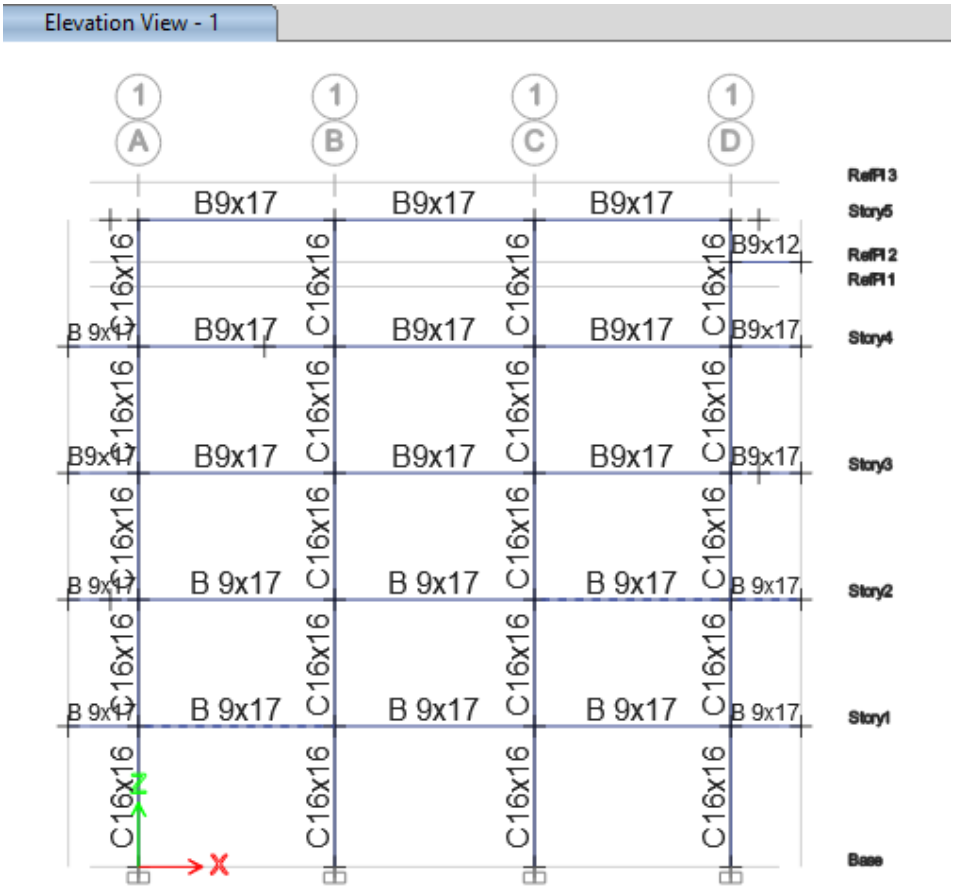


Figure 3-3: Elevation along grid 2-2 of Model created in Etabs

4 STRUCTURAL ANALYSIS

For the purpose of structural analysis IS Codes are followed for loadings, and Nepal Building Code is considered for load combinations and for other analysis procedures are followed.

4.1 LOAD CASES CONSIDERED

Following loads have been considered in the analysis of the building as per NBC 105:2025.

1. Dead Load (DL)
2. Live load (LL)
3. Earthquake load in +ve X-direction (EQ_X)
4. Earthquake load in –ve X-direction ($-EQ_X$)
5. Earthquake load in +ve Y-direction (EQ_Y)
6. Earthquake load in –ve Y-direction ($-EQ_Y$)

4.2 LOAD COMBINATION

Following load combinations have been adopted as per NBC 105:2025 for Static Analysis.

For Parallel System

1.2DL+1.5LL

$DL+\lambda LL \pm EQ_{(x,y)}$

$0.9DL \pm EQ_{(x,y)}$

Where,

$\lambda = 0.6$ for storage facilities.

$= 0.3$ for other usage

DL= Dead load

LL = Imposed (Live) load

$EQ_{(x,y)}$ = Earthquake load (Along X and Y direction)

For Non-Parallel System

1.2DL+1.5LL

$DL+\lambda LL \pm (EQ_x \pm 0.3EQ_y)$

$DL+\lambda LL \pm (EQ_y \pm 0.3EQ_x)$

$0.9DL \pm (EQ_x \pm 0.3EQ_y)$

$0.9DL \pm (EQ_y \pm 0.3EQ_x)$

Where,

$\lambda = 0.6$ for storage facilities.

$= 0.3$ for other usage

DL= Dead load

LL = Imposed (Live) load

$EQ_{(x,y)}$ = Earthquake load (Along X and Y direction)

4.3 DEAD LOADS

Dead loads are assumed to be produced by slab, beams, columns, walls, parapet walls, plasters and mortars and floor finish. The weight of building materials are taken as per IS 875 (Part 1-1987).

Table 4-1: Dead Loads

Specific weight of materials [Ref: IS: 875(Part 1)-1987)]	
Materials	Unit weight(γ)
Reinforced Concrete	25.00 kN/m³
Brick Masonry	19.20 kN/m³
Floor Finishing (Screeding & punning)	23.00 kN/m³
Cement Sand Plaster	20.40 kN/m³

Wall Loads

The wall loads are applied to the associated beams as uniformly distributed loads. Reduction for the openings is made wherever necessary.

UDL due to wall = height * thickness * sp. Weight

Table 4-2: Wall loads Applied to the Frames

S.N.	Wall t(mm)	Storey Ht(m)	Beam d(mm)	Ht. of Wall (m)	kN/m	Opening	kN/m	Adopt UDL	Remarks
1	230	2.8448	355.6	2.4892	11.450	0%	11.450	12.	230mm wall below Beam
	230	2.8448	355.6	2.4892	11.450	20%	9.160	10	230mm wall below Beam
	230	2.8448	355.6	2.4892	11.450	30%	8.015	9	230mm wall below Beam
2	127	2.8448	355.6	2.4892	6.323	0%	6.323	7	127 mm wall below Beam
	127	2.8448	355.6	2.4892	6.323	20%	5.058	6	127 mm wall below Beam
	127	2.8448	355.6	2.4892	6.323	30%	4.426	5	127 mm wall below Beam

4.4 LIVE LOADS

Live loads are applied on floor slabs on the basis of usage of rooms, as specified in IS 875 part II.

Table 4-3: Live Loads

Rooms	2.0 kN/m^2
Toilet/ Bathrooms	2.0 kN/m^2

Staircase/Balcony	3.0 kN/m^2
Roof	1.5 kN/m^2

4.5 LOADS ON SUPER-STRUCTURE

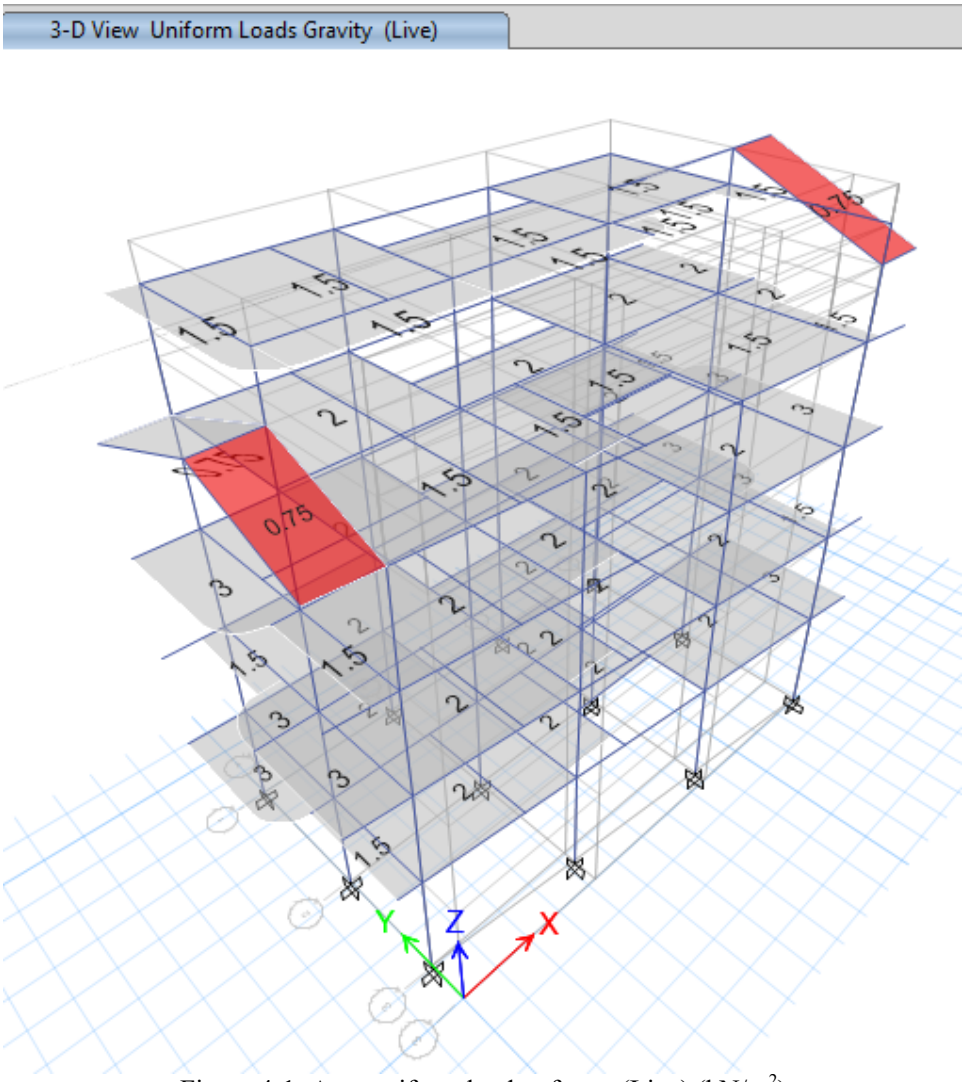


Figure 4-1: Area uniform load to frame (Live) (kN/m^2)

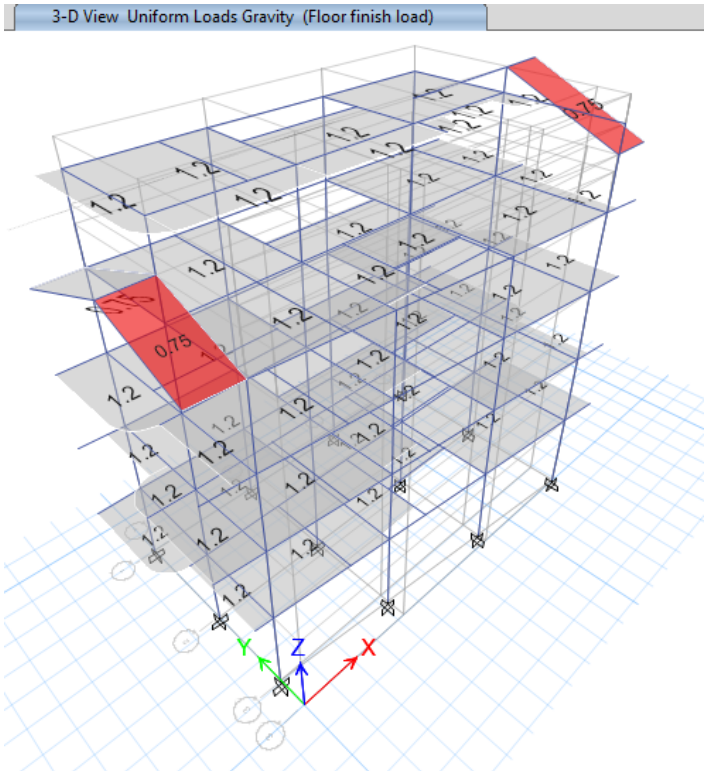


Figure 4-2: Area uniform load to frame (Floor finish) (kN/m^2)

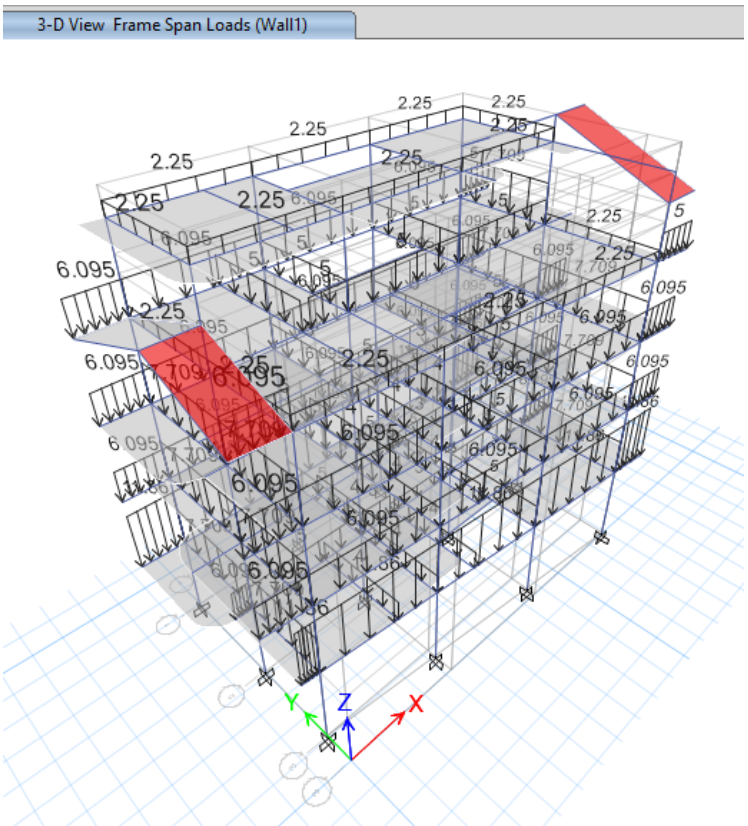


Figure 4-3: Wall loads on beams (kN/m)

4.6 SEISMIC LOADS (EARTHQUAKE LOADS)

4.6.1 Seismic Coefficient Method

Sesmic Coefficient As Per NBC 105:2025			
Height of Building=		13.7153307	45
SesmicZoining Factor (Z)=		0.35	
Important Factor=		1	
Soil Type		D	
Period of vibration:			
For Reinforced Moment Resisting Frame kt =		0.075	
		0.53452248	
T1 =1.25* kt H ¾		0.6681531	
Lower Period of the flat part of the Spectrum Ta=		0.5	
Upper period of the flat part of the Spectrum Tc =		2	2.670383
Peak Spectral Acceleration Normalized by PGAo α =		2.25	2.25
Coefficient to control the decending branch of Spectrum K=		0.8	52.25472
Calculation of Spectral Shape Factor, Ch(T)		2.25	
Since,			
'=$\alpha if Ta \le T \le Tc$			
Ch(T)	=	2.25	
Elastic site spectra for horizontal loading (Clause 4.1.1 NBC 105:2025)			
C (T) =Ch(T) Z I	=	0.7875	
elastic site spectra for Serviceability Limit State(Clause 4.2 NBC 105:2025)			
mm **Cs (T) = 0.20 C (T)**	=	0.1575	
horizontal base shear coefficient for Equivalent Static Method :			
Ref. Table 5.2 NBC 105:2020			
Ductility Factor for ULS **(Ru)**=	=	4	
Over-strength factor for ULS **(Ωu)**=	=	1.5	
Over-strength factor for SLS **(Ωs)**=	=	1.25	
Ultimate limit state			
horizontal base shear coefficient at the ULS State (Clause 6.1(1) NBC 105:2025)			
Cd(T1)=C(T1)/(Rμ x Ωu)	=	0.13125	
Serviceability limit state,			
horizontal base shear coefficient at the SLS State (Clause 6.1(2) NBC 105:2025)			
Cd(T1)=Cs(T1)/Ωs	=	0.126	
exponent related to the structural period (k)	=	1.08407655	
Displacement Criteria for Eqivalent Static Method:			
Alloable ratio ULS 0.025	=	0.025	

Allowable Displacement ULS	=0.025*H/Rμ		=	85.7208168	mm
Alloable ratio SLS		0.006	=	0.006	
Allowable Displacement SLS	=0.006*H/Rs		=	82.2919842	
DriftCriteria for Eqivalent Static Method:					
Story drift from ETABS ULS			=	0.0057	OK
Allowable drift ULS	=0.025/Rμ		=	0.00625	
Story drift from ETABS SLS			=	0.0055	OK
Allowable Drift SLS	=0.006/Rs		=	0.006	

4.6.2 Dynamic Analysis

Response Spectrum Method

The dynamic analysis was performed by Response spectrum Method using a design spectrum offered by ***NBC 105: 2025.***

For the purpose of analysis, seismic forces are applied in the model of the building in Etabs 2019.1.0. Hence, the manual calculations of seismic weight, base shear and the seismic forces have not been shown. However, the Etabs 2019.1.0 outputs for the Seismic Weight, Base Shear are shown.

Cracked Section Analysis

The effective stiffness of Different section shall be taken as: (NBC 105:2025 cl.3.4 Table 3-1)

S No.	Component	Flexural Stiffness	
		Ultimate	Service
1	Beam	0.35 Ec Ig	0.7 Ec Ig
2	Column	0.7 Ec Ig	0.9 Ec Ig
3	Wall-cracked	0.5 Ec Ig	0.7 Ec Ig
4	Wall-uncracked	0.6 Ec Ig	0.9 Ec Ig
5	Masonary Wall	0.45 Ec Ig	0.6 Ec Ig
6	Flat Slabs	0.45 Ec Ig	0.5 Ec Ig

4.7 AUTO SEISMIC USER COEFFICIENT TABLE

Name	Is Auto Load	Ecc Ratio	Top Story	Bottom Story	C	K	Weight Used kN	Base Shear kN
Eq-X(SLS)	No	0.05	Story5	Base	0.126	1.08		
Eq-X(SLS)(1/3)	Yes	0.05	Story5	Base	0.126	1.08	6060.3898	763.6091
Eq-X(SLS)(2/3)	Yes	0.05	Story5	Base	0.126	1.08	6060.3898	763.6091
Eq-X(SLS)(3/3)	Yes	0.05	Story5	Base	0.126	1.08	6060.3898	763.6091
Eq-X(ULS)	No	0.05	Story5	Base	0.131	1.08		
Eq-X(ULS)(1/3)	Yes	0.05	Story5	Base	0.131	1.08	6060.3898	793.9111

Eq-X(ULS)(2/3)	Yes	0.05	Story5	Base	0.131	1.08	6060.3898	793.9111
Eq-X(ULS)(3/3)	Yes	0.05	Story5	Base	0.131	1.08	6060.3898	793.9111
Eq-Y(SLS)	No	0.05	Story5	Base	0.126	1.08		
Eq-Y(SLS)(1/3)	Yes	0.05	Story5	Base	0.126	1.08	6060.3898	763.6091
Eq-Y(SLS)(2/3)	Yes	0.05	Story5	Base	0.126	1.08	6060.3898	763.6091
Eq-Y(SLS)(3/3)	Yes	0.05	Story5	Base	0.126	1.08	6060.3898	763.6091
Eq-Y(ULS)	No	0.05	Story5	Base	0.131	1.08		
Eq-Y(ULS)(1/3)	Yes	0.05	Story5	Base	0.131	1.08	6060.3898	793.9111
Eq-Y(ULS)(2/3)	Yes	0.05	Story5	Base	0.131	1.08	6060.3898	793.9111
Eq-Y(ULS)(3/3)	Yes	0.05	Story5	Base	0.131	1.08	6060.3898	793.9111

4.8 MODAL ANALYSIS AND MODAL MASS PARTICIPATION

No of modes considered for Modal Analysis:50
Mass Participation: More than 90% in both directions(NBC 105:2025 cl.7.3)

Table 4-4: Modal Mass Participation Ratio

Case	Mode	Period sec	UX	UY	UZ	SumUX	SumUY	SumUZ
Modal	1	0.933	0.5606	0.1501	0	0.5606	0.1501	0
Modal	2	0.909	0.2193	0.5843	0	0.7799	0.7344	0
Modal	3	0.773	0.0468	0.1017	0	0.8267	0.8361	0
Modal	4	0.307	0.1111	0.0001	0	0.9378	0.8361	0
Modal	5	0.286	0.0007	0.1008	0	0.9385	0.937	0
Modal	6	0.249	0.0019	0.0146	0	0.9404	0.9516	0
Modal	7	0.17	0.0401	5.414E-07	0	0.9805	0.9516	0
Modal	8	0.153	1.145E-05	0.0328	0	0.9805	0.9844	0
Modal	9	0.136	1.05E-05	0.0027	0	0.9805	0.9871	0
Modal	10	0.112	0.0162	1.615E-05	0	0.9967	0.9871	0
Modal	11	0.105	1.387E-05	0.0097	0	0.9967	0.9967	0
Modal	12	0.098	5.084E-07	0.0002	0	0.9967	0.9969	0

4.9 DESIGN ECCENTRICITY

Esi= CM-CR
where,
Edi=Design Eccentricity, Esi= Static Eccentricity= Distance e Between CM and CR
CM= Center of Mass, CR= Center of Rigidity
Bi = Dimension of Building along the direction considered

Building Dimension Along X (Bx), for first floor & typical floor= 15.93m
Building Dimension Along Y for first floor & typical floor(By=8.46m
Building Dimension Along X for top floor (Bx)=11.81 m
Building Dimension Along Y top floor (By)=8.46 m

Table 4-5: Design Eccentricity

										Check	
Story	**Diaphragm**	**XCCM**	**YCCM**	**XCR**	**YCR**	**ex**	**ey**	**Lx**	**Ly**	**x**	**y**
		m	m	m	m					**x**	**y**
Story5	D1	6.61	5.61	6.05	4.38	0.56	1.23	15.93	8.10	0.035	0.152
Story4	D1	6.96	5.07	6.16	4.63	0.80	0.43	15.93	8.10	0.050	0.053
Story3	D1	6.81	4.76	6.21	4.36	0.60	0.40	15.93	8.10	0.038	0.049
Story2	D1	6.76	4.64	6.23	4.29	0.53	0.35	15.93	8.10	0.033	0.043
Story1	D1	6.52	4.42	6.31	4.19	0.21	0.24	15.93	8.10	0.013	0.029

Hence, Building eccentricity has been found within 10%. Therefore, dynamic analysis is not mandatory.

4.10 CHECK FOR TORSIONAL IRREGULARITY

Torsion Irregularity is checked as per (NBC 105:2025 cl.5.5.2.1)
If $\Delta_{max} > 1.5\ \Delta_{min}$, building is considered to have torsional irregularity.
Where,
Δ_{max} = Maximum displacement of the Storey under the action of Earthquake load
Δ_{min} = Minimum displacement of the Storey under the action of Earthquake load.

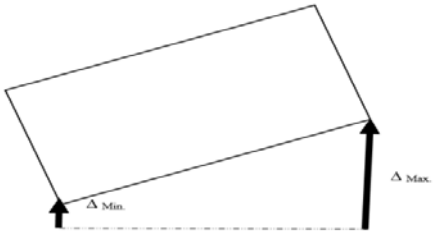


Table 4-6 Ratio of Maximum to Average Storey Displacements for different Storey

TABLE: Story Max Over Avg Displacement

Story	Output Case	Step Type	Step Number	Direction	Maximum mm	Average mm	Ratio
Story5	Modal	Mode	1	X	0.06	0.06	1.13
Story5	Eq-X(SLS)	Step By Step	1	X	46.87	45.77	1.02
Story4	Eq-X(SLS)	Step By Step	1	X	41.90	39.60	1.06
Story3	Eq-X(SLS)	Step By Step	1	X	33.45	31.67	1.06
Story2	Eq-X(SLS)	Step By Step	1	X	22.23	21.08	1.06
Story1	Eq-X(SLS)	Step By Step	1	X	9.82	9.32	1.05
Story5	Eq-X(SLS)	Step By Step	2	X	48.32	46.49	1.04

Story4	Eq-X(SLS)	Step By Step	2	X	43.29	39.51	1.10
Story3	Eq-X(SLS)	Step By Step	2	X	34.60	31.60	1.10
Story2	Eq-X(SLS)	Step By Step	2	X	23.01	21.03	1.09
Story1	Eq-X(SLS)	Step By Step	2	X	10.18	9.30	1.09
Story5	Eq-X(SLS)	Step By Step	3	X	45.43	45.06	1.01
Story4	Eq-X(SLS)	Step By Step	3	X	40.52	39.69	1.02
Story3	Eq-X(SLS)	Step By Step	3	X	32.30	31.74	1.02
Story2	Eq-X(SLS)	Step By Step	3	X	21.45	21.12	1.02
Story1	Eq-X(SLS)	Step By Step	3	X	9.47	9.34	1.01
Story5	Eq-X(ULS)	Step By Step	1	X	48.74	47.59	1.02
Story4	Eq-X(ULS)	Step By Step	1	X	43.56	41.17	1.06
Story3	Eq-X(ULS)	Step By Step	1	X	34.78	32.92	1.06
Story2	Eq-X(ULS)	Step By Step	1	X	23.11	21.91	1.06
Story1	Eq-X(ULS)	Step By Step	1	X	10.21	9.69	1.05
Story5	Eq-X(ULS)	Step By Step	2	X	50.23	48.33	1.04
Story4	Eq-X(ULS)	Step By Step	2	X	45.00	41.08	1.10
Story3	Eq-X(ULS)	Step By Step	2	X	35.97	32.85	1.10
Story2	Eq-X(ULS)	Step By Step	2	X	23.93	21.87	1.09
Story1	Eq-X(ULS)	Step By Step	2	X	10.58	9.67	1.09
Story5	Eq-X(ULS)	Step By Step	3	X	47.24	46.85	1.01
Story4	Eq-X(ULS)	Step By Step	3	X	42.13	41.26	1.02
Story3	Eq-X(ULS)	Step By Step	3	X	33.58	33.00	1.02
Story2	Eq-X(ULS)	Step By Step	3	X	22.30	21.96	1.02
Story1	Eq-X(ULS)	Step By Step	3	X	9.85	9.71	1.01
Story5	Eq-Y(SLS)	Step By Step	1	Y	46.10	40.76	1.13
Story4	Eq-Y(SLS)	Step By Step	1	Y	42.75	38.04	1.12
Story3	Eq-Y(SLS)	Step By Step	1	Y	34.57	30.97	1.12
Story2	Eq-Y(SLS)	Step By Step	1	Y	23.01	20.69	1.11
Story1	Eq-Y(SLS)	Step By Step	1	Y	10.04	9.13	1.10
Story5	Eq-Y(SLS)	Step By Step	2	Y	51.65	41.06	1.26
Story4	Eq-Y(SLS)	Step By Step	2	Y	47.97	38.28	1.25
Story3	Eq-Y(SLS)	Step By Step	2	Y	38.86	31.16	1.25
Story2	Eq-Y(SLS)	Step By Step	2	Y	25.92	20.82	1.25
Story1	Eq-Y(SLS)	Step By Step	2	Y	11.33	9.18	1.24
Story5	Eq-Y(SLS)	Step By Step	3	Y	40.54	40.47	1.00
Story4	Eq-Y(SLS)	Step By Step	3	Y	38.07	37.80	1.01
Story3	Eq-Y(SLS)	Step By Step	3	Y	31.29	30.78	1.02
Story2	Eq-Y(SLS)	Step By Step	3	Y	21.03	20.57	1.02
Story1	Eq-Y(SLS)	Step By Step	3	Y	9.43	9.09	1.04
Story5	Eq-Y(ULS)	Step By Step	1	Y	47.93	42.38	1.13
Story4	Eq-Y(ULS)	Step By Step	1	Y	44.45	39.55	1.12
Story3	Eq-Y(ULS)	Step By Step	1	Y	35.94	32.20	1.12
Story2	Eq-Y(ULS)	Step By Step	1	Y	23.93	21.51	1.11
Story1	Eq-Y(ULS)	Step By Step	1	Y	10.44	9.50	1.10
Story5	Eq-Y(ULS)	Step By Step	2	Y	53.70	42.69	1.26
Story4	Eq-Y(ULS)	Step By Step	2	Y	49.88	39.80	1.25
Story3	Eq-Y(ULS)	Step By Step	2	Y	40.40	32.40	1.25
Story2	Eq-Y(ULS)	Step By Step	2	Y	26.94	21.64	1.25
Story1	Eq-Y(ULS)	Step By Step	2	Y	11.78	9.54	1.24

Story5	Eq-Y(ULS)	Step By Step	3	Y	42.15	42.08	1.00
Story4	Eq-Y(ULS)	Step By Step	3	Y	39.58	39.30	1.01
Story3	Eq-Y(ULS)	Step By Step	3	Y	32.53	32.01	1.02
Story2	Eq-Y(ULS)	Step By Step	3	Y	21.86	21.39	1.02
Story1	Eq-Y(ULS)	Step By Step	3	Y	9.80	9.45	1.04

The table shows that the ratio $\Delta_{max}/\Delta_{min}$ has been limited to limiting value of 1.5, hence indicating that Torsional Irregularities does not exist in building.

4.11 ANALYSIS &DESIGN PARAMETERS

Followings factors are considered for earthquake resistant design of the building.

Table 4-7: Analysis & Design Parameters

Method adopted for design		Seismic Coefficient method
Design code		NBC 105 :2025
Height of building		14.05 m
Zone factor(Z)		0.35
Soil Type		Type D
Importance factor(I)		1
Ductility factor (Rμ)		4
Over strength factor(Ωu)		1.5
Over-strength factor for SLS (Ωs)		1.25
Mass participation in dynamic analysis		Above 90 % along both directions
Fundamental Time Period (T)	Along X (sec)	0.54282181
	Along Y (sec)	0.54282181
Seismic Weight (W) (kN)		6060.3898
Lower period of flat part (Ta) (sec)		0.5
Upper period of flat part (Tc) (sec)		2
Spectral shape factor (Ch(t))		2.25
Elastic site spectra ULS (C(t))		0.7875
Elastic site spectra SLS (C(t))		0.1575
Horizontal base shear coefficient ULS		0.13125
Horizontal base shear coefficient SLS		0.126
Exponent Related to structural period K		1.08
Base shear from seismic coefficient method along –x (KN)		793.9111
Base shear from seismic coefficient method along -y (KN)		793.9111

4.12 STORY DRIFT CALCULATIONS

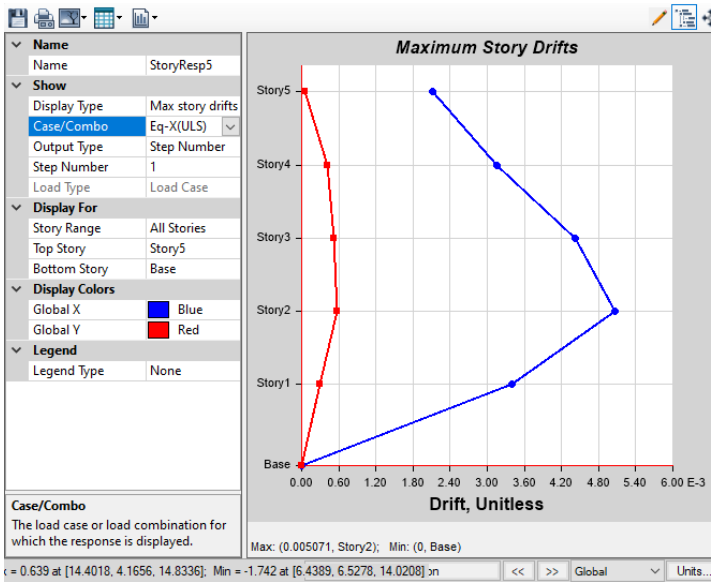
Table 4-8: Story Drift against Earthquake Loads

Story	Output Case	Step Type	Step Number	Direction	Max Drift mm	Avg Drift mm	Ratio
Story5	Modal	Mode	1	X	0.01	0.01	1.12
Story5	Eq-X(SLS)	Step By Step	1	X	5.69	5.32	1.07
Story5	Eq-X(SLS)	Step By Step	2	X	5.58	5.26	1.06
Story5	Eq-X(SLS)	Step By Step	3	X	5.80	5.36	1.08

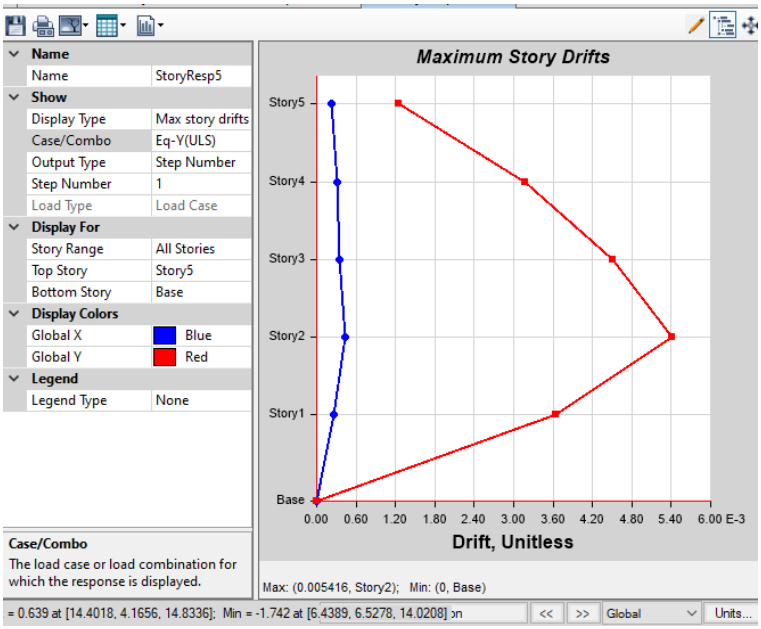
Story5	Eq-X(ULS)	Step By Step	1	X	5.92	5.53	1.07
Story5	Eq-X(ULS)	Step By Step	2	X	5.80	5.47	1.06
Story5	Eq-X(ULS)	Step By Step	3	X	6.03	5.57	1.08
Story5	Eq-Y(SLS)	Step By Step	1	Y	3.40	2.75	1.24
Story5	Eq-Y(SLS)	Step By Step	2	Y	3.74	2.80	1.33
Story5	Eq-Y(SLS)	Step By Step	3	Y	3.06	2.70	1.14
Story5	Eq-Y(ULS)	Step By Step	1	Y	3.54	2.86	1.24
Story5	Eq-Y(ULS)	Step By Step	2	Y	3.89	2.92	1.33
Story5	Eq-Y(ULS)	Step By Step	3	Y	3.18	2.80	1.14
Story4	Eq-X(SLS)	Step By Step	1	X	8.45	7.93	1.07
Story4	Eq-X(SLS)	Step By Step	2	X	8.69	7.91	1.10
Story4	Eq-X(SLS)	Step By Step	3	X	8.22	7.95	1.03
Story4	Eq-X(ULS)	Step By Step	1	X	8.79	8.24	1.07
Story4	Eq-X(ULS)	Step By Step	2	X	9.03	8.22	1.10
Story4	Eq-X(ULS)	Step By Step	3	X	8.55	8.26	1.03
Story4	Eq-Y(SLS)	Step By Step	1	Y	8.59	7.11	1.21
Story4	Eq-Y(SLS)	Step By Step	2	Y	9.72	7.16	1.36
Story4	Eq-Y(SLS)	Step By Step	3	Y	7.46	7.06	1.06
Story4	Eq-Y(ULS)	Step By Step	1	Y	8.93	7.39	1.21
Story4	Eq-Y(ULS)	Step By Step	2	Y	10.11	7.45	1.36
Story4	Eq-Y(ULS)	Step By Step	3	Y	7.75	7.34	1.06
Story3	Eq-X(SLS)	Step By Step	1	X	11.22	10.58	1.06
Story3	Eq-X(SLS)	Step By Step	2	X	11.59	10.55	1.10
Story3	Eq-X(SLS)	Step By Step	3	X	10.85	10.61	1.02
Story3	Eq-X(ULS)	Step By Step	1	X	11.67	11.00	1.06
Story3	Eq-X(ULS)	Step By Step	2	X	12.05	10.97	1.10
Story3	Eq-X(ULS)	Step By Step	3	X	11.29	11.03	1.02
Story3	Eq-Y(SLS)	Step By Step	1	Y	11.93	10.30	1.16
Story3	Eq-Y(SLS)	Step By Step	2	Y	13.63	10.36	1.32
Story3	Eq-Y(SLS)	Step By Step	3	Y	10.34	10.24	1.01
Story3	Eq-Y(ULS)	Step By Step	1	Y	12.40	10.71	1.16
Story3	Eq-Y(ULS)	Step By Step	2	Y	14.17	10.78	1.32
Story3	Eq-Y(ULS)	Step By Step	3	Y	10.76	10.64	1.01
Story2	Eq-X(SLS)	Step By Step	1	X	12.43	11.76	1.06
Story2	Eq-X(SLS)	Step By Step	2	X	12.86	11.73	1.10
Story2	Eq-X(SLS)	Step By Step	3	X	12.00	11.79	1.02
Story2	Eq-X(ULS)	Step By Step	1	X	12.92	12.22	1.06
Story2	Eq-X(ULS)	Step By Step	2	X	13.37	12.19	1.10
Story2	Eq-X(ULS)	Step By Step	3	X	12.47	12.26	1.02
Story2	Eq-Y(SLS)	Step By Step	1	Y	13.32	11.58	1.15
Story2	Eq-Y(SLS)	Step By Step	2	Y	15.30	11.66	1.31
Story2	Eq-Y(SLS)	Step By Step	3	Y	11.65	11.43	1.02
Story2	Eq-Y(ULS)	Step By Step	1	Y	13.85	12.04	1.15
Story2	Eq-Y(ULS)	Step By Step	2	Y	15.91	12.12	1.31
Story2	Eq-Y(ULS)	Step By Step	3	Y	12.11	11.89	1.02
Story1	Eq-X(SLS)	Step By Step	1	X	9.82	9.32	1.05
Story1	Eq-X(SLS)	Step By Step	2	X	10.18	9.30	1.09
Story1	Eq-X(SLS)	Step By Step	3	X	9.47	9.34	1.01
Story1	Eq-X(ULS)	Step By Step	1	X	10.21	9.69	1.05

Story1	Eq-X(ULS)	Step By Step	2	X	10.58	9.67	1.09
Story1	Eq-X(ULS)	Step By Step	3	X	9.85	9.71	1.01
Story1	Eq-Y(SLS)	Step By Step	1	Y	10.04	9.13	1.10
Story1	Eq-Y(SLS)	Step By Step	2	Y	11.33	9.18	1.24
Story1	Eq-Y(SLS)	Step By Step	3	Y	9.43	9.09	1.04
Story1	Eq-Y(ULS)	Step By Step	1	Y	10.44	9.50	1.10
Story1	Eq-Y(ULS)	Step By Step	2	Y	11.78	9.54	1.24
Story1	Eq-Y(ULS)	Step By Step	3	Y	9.80	9.45	1.04

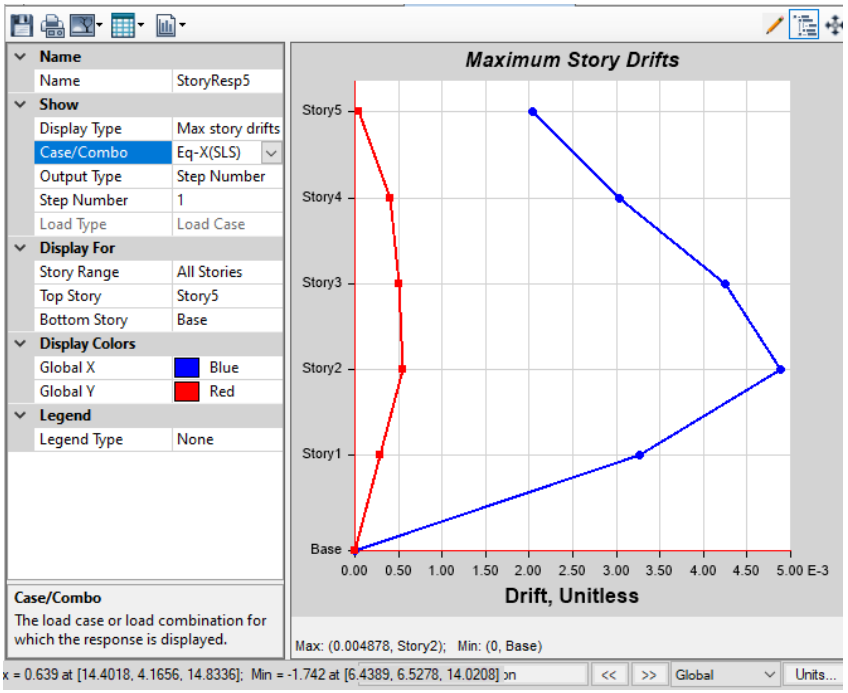
It is seen that the max drift ratios are:



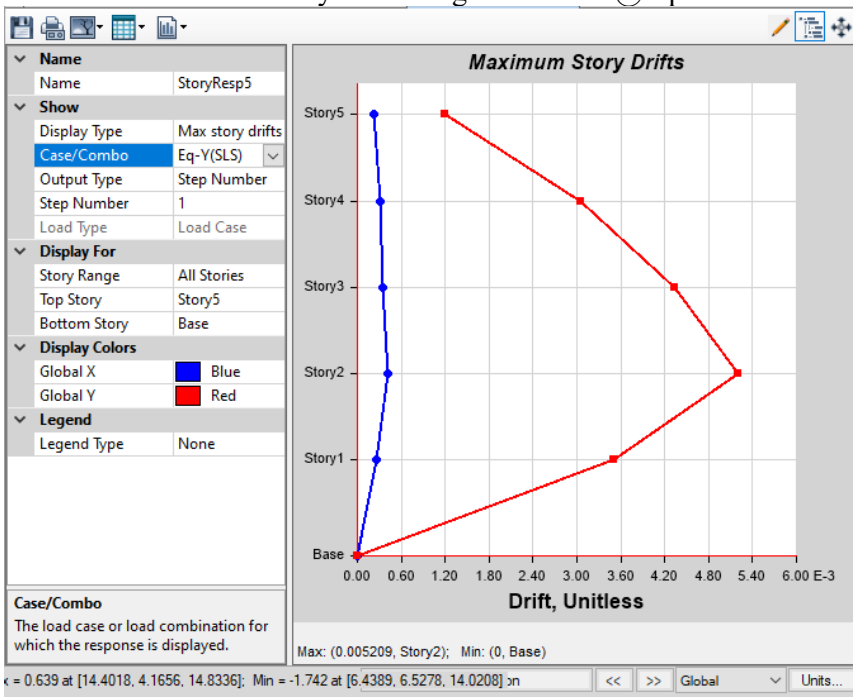
Maximum storey drift along x-direction @EqxUL



Maximum storey drift along Y-direction @EqyUL



Maximum storey drift along x-direction @EqxSL



Maximum storey drift along Y-direction @EqySL

At Ultimate limit state:

-At cracked section:

Maximum drift ratio at cracked section 0.0025 (NBC 105:2025 cl.5.5.3)

At Serviceability limit state:

-At cracked section:

Maximum drift ratio at cracked section 0.006

Hence, Drifts are within the safe limits (NBC 105:2025 cl.5.5.3)

4.13 MOMENTS AND SHEAR FORCES IN FRAMES

The moments and shear forces of the buildings were analyzed using the Etabs 2019. The analysis is used for identification of critical sections and to find out the design requirements so as to design various structural components. The sample moment diagrams and shear force diagram samples of the sections around these critical components, as extracted from SAP 2000 are as follows:

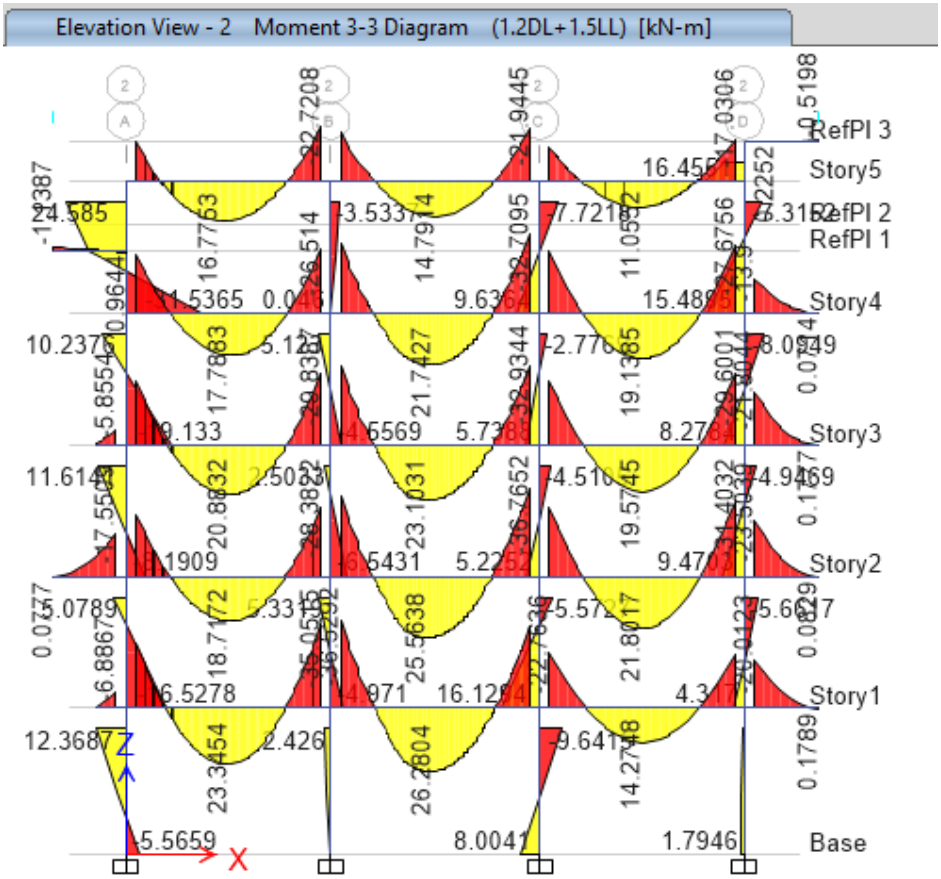


Figure 4-4: Sample Bending Moment Diagram for frames
Along Grid 2-2(1.2DL+1.5LL) Unit; kNm

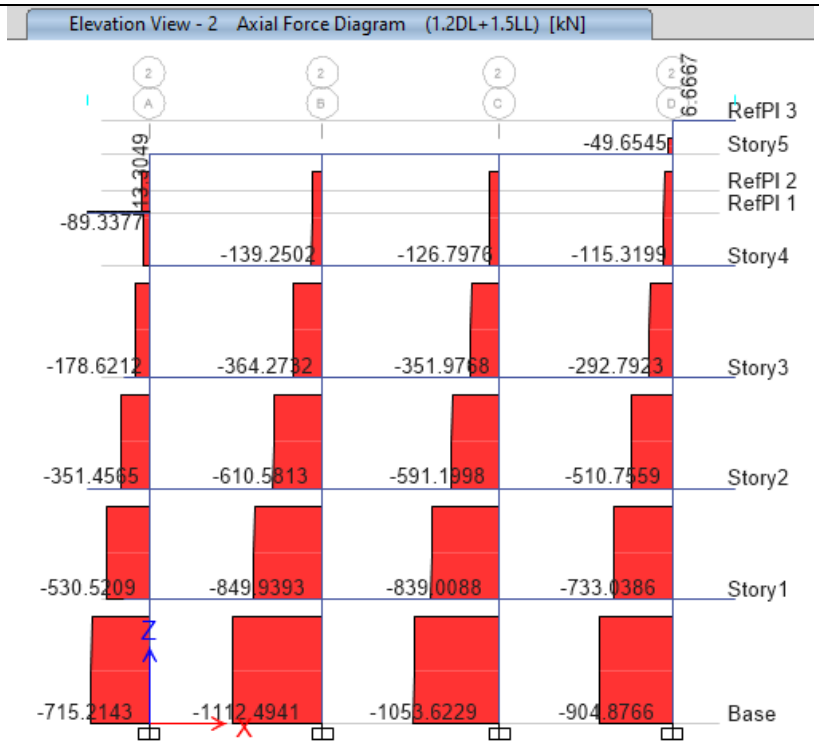


Figure 4-5: Sample Axial Force Diagram for frames
Along Grid 2-2(1.2DL+1.5LL) Unit; kN

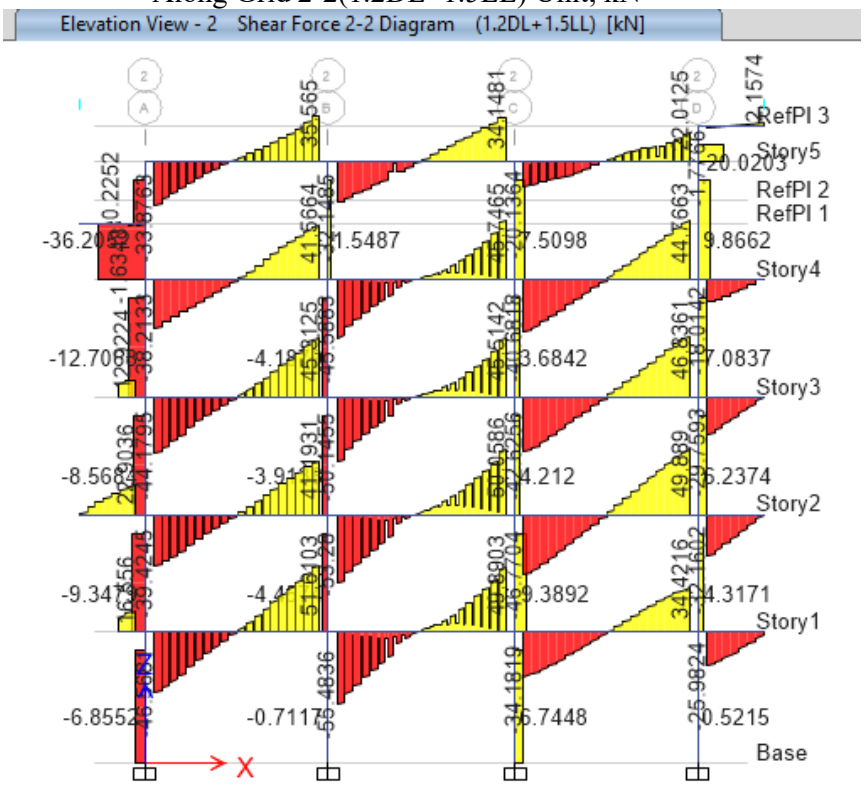


Figure 4-6: Sample Shear Diagram for frames
Along Grid 2-2(1.2DL+1.5LL) Unit; kNm

`DESIGN

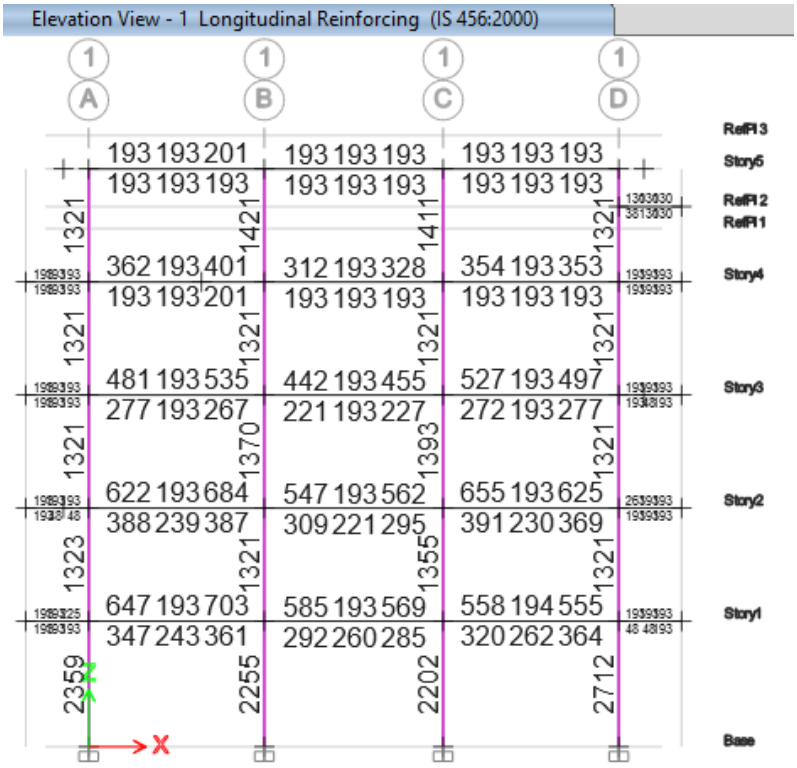
The design of reinforced concrete structural members includes selection of material properties (grade of steel and concrete), shape and size of cross section, factor of safety and amount of steel required. The design of reinforced concrete members are carried out using limit state method as per IS 456: 2000. The limit state method is the modern and latest design methodology. This method evolved around 1970's. Limit state method is based on the concept of multiple safety factors and attempts to provide adequate safety at the ultimate loads and adequate serviceability at service loads.

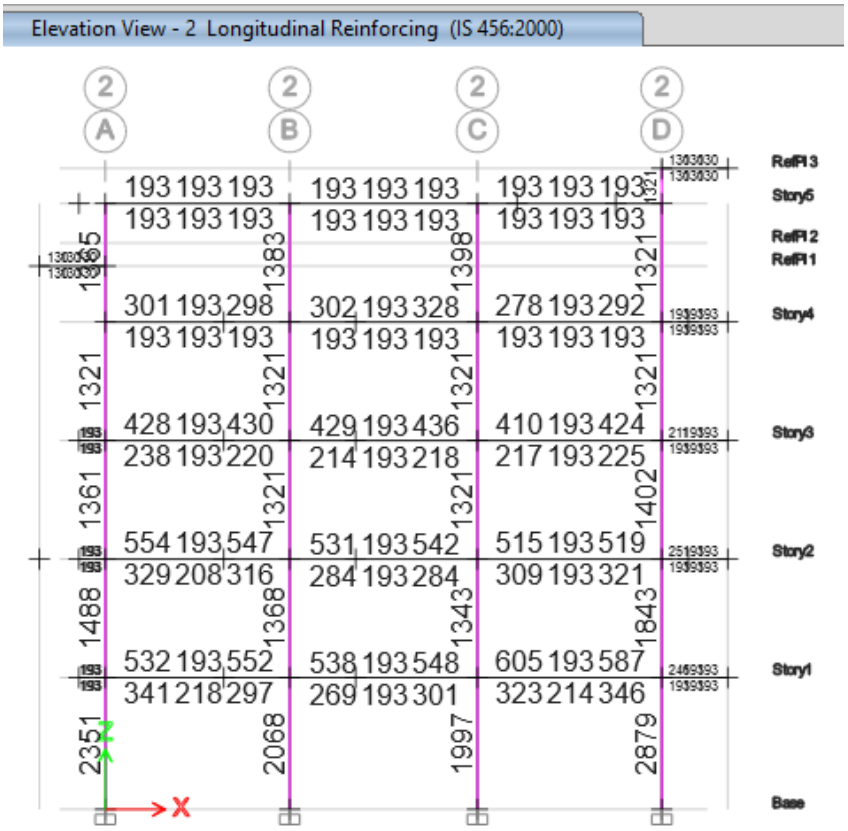
For the design of the members, IS 456:2000 and design aid SP 16 has been used. Footings have been checked for vertical loads and moments developed at the base due to dead load and live load only. Square footings have been adopted from seismic point of view that reversal stress may occur. And footing beams are provided for column at foundation for more rigidity of building and also need for the column located at boundary. Longitudinal reinforcement in beams and columns has been calculated based on critical load combination. Spacing of the shear reinforcement has been calculated as per the ductility criteria as defined in IS 13920 - 2016.Some sample designs are shown later on in this report.;

Beams and columns have been designed using Etabs 2019.1.0. While slab, staircase are designed manually and some foundations are designed manually and some are using SAFEv16. Samples of manual design calculation of critical beam, column slab and footings are shown in this report in. **The structural design of sections and reinforcements are presented in the drawing.**

4.14 DESIGN SUMMARY

Column Design Summary




Elevation View - 2 Longitudinal Reinforcing (IS 456:2000)
2 A
2 B
2 C
2 D
RefFl 3
Story5
RefFl 2
RefFl 1
Story4
Story3
Story2
Story1
Base
193 193 193
301 193 298
428 193 430
554 193 547
329 208 316
532 193 552
341 218 297
193 193 193
302 193 328
429 193 436
531 193 542
284 193 284
538 193 548
269 193 301
193 193 193
278 193 292
410 193 424
515 193 519
309 193 321
605 193 587
323 214 346
X

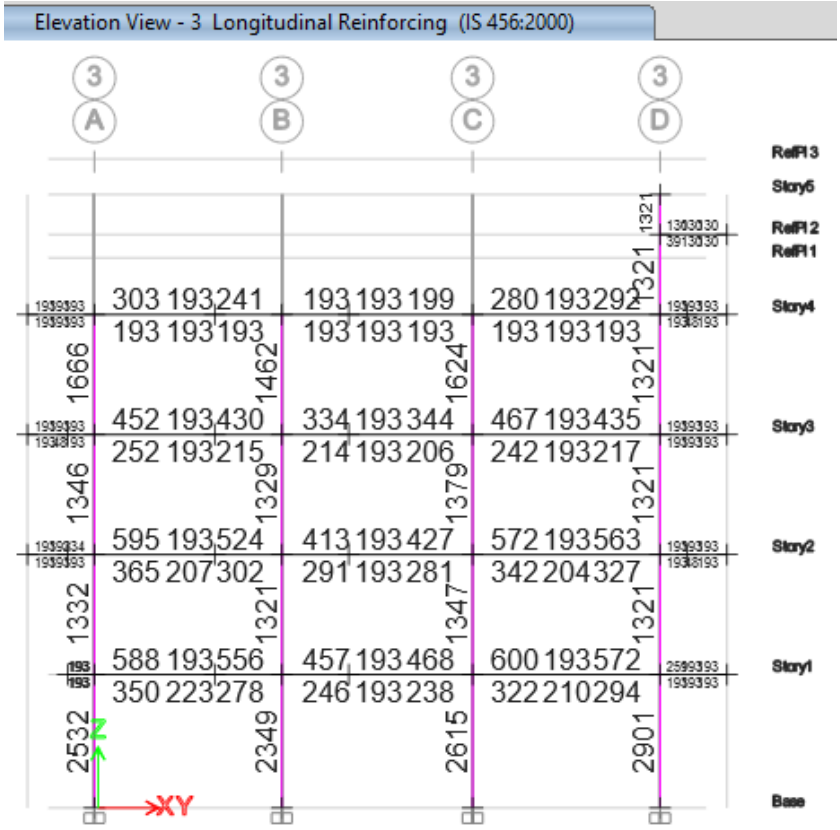

Elevation View - 3 Longitudinal Reinforcing (IS 456:2000)
3 A
3 B
3 C
3 D
RefFl 3
Story5
RefFl 2
RefFl 1
Story4
Story3
Story2
Story1
Base
303 193 241
452 193 430
595 193 524
365 207 302
588 193 556
350 223 278
193 193 199
334 193 344
413 193 427
291 193 281
457 193 468
246 193 238
280 193 292
467 193 435
572 193 563
342 204 327
600 193 572
322 210 294
193 193 193
252 193 215
214 193 206
242 193 217
XY

Table 0-1: Column Design Summary

COLUMN REINFORCEMENT DETAILS

Concrete Grade: M25

Steel Grade: Fe500

Storey Lev.	Colm ID	Column	Width b	Depth D	Ast from ETABS	No of 20ϕ	No of 16ϕ	No of 12ϕ	Tot Ast prov mm²	Tot Ast prov %	Stirrups	Ast%	Remark
			mm	mm	mm²								
Top FL	C1	D1, D2 & D3	400	400	1321		8	4	2061	1.56	8ø @4" for H/6 or atleast lateral dimension or 1'6" whichever is greater & 8ø @6"c/c at remaining height	1.29	
Third FL			400	400	1328		12		2413	1.82		1.51	
Second FL			400	400	1438	4	8		2865	1.99		1.79	
First FL			400	400	2185	8	4		3318	1.52		2.07	
Ground Floor			400	400	2789	8	4		3318	1.19		2.07	
Top FL	C2	B1, B2, C1 & C2	400	400	1445		8	4	2061	1.43	8ø @4" for H/6 or atleast lateral dimension or 1'6" whichever is greater & 8ø @6"c/c at remaining height	1.29	
Third FL			400	400	1321		8	4	2061	1.56		1.29	
Second FL			400	400	1818		12		2413	1.33		1.51	
First FL			400	400	1484	4	8		2865	1.93		1.79	
Ground Floor			400	400	2436	8	4		3318	1.36		2.07	
Top FL	C3	A1 & A2	400	400	1358		4	8	1709	1.26	8ø @4" for H/6 or atleast lateral dimension or 1'6" whichever is greater & 8ø @6"c/c at remaining height	1.07	
Third FL			400	400	1336		8	4	2061	1.54		1.29	
Second FL			400	400	1468		12		2413	1.64		1.51	
First FL			400	400	1886	4	8		2865	1.52		1.79	
Ground Floor			400	400	2359	4	8		2865	1.21		1.79	
Top FL	C4a	C3	---	---	-				0	######	8ø @4" for H/6 or atleast lateral dimension or 1'6" whichever is greater & 8ø @6"c/c at remaining height	####	
Third FL			400	400	1856		8	4	2061	1.11		1.29	
Second FL			400	400	1451		8	4	2061	1.42		1.29	
First FL			400	400	1449		12		2413	1.67		1.51	
Ground Floor			400	400	2615	4	8		2865	1.10		1.79	
Top FL	C4	A3 & B3	---	---	-				0	######	8ø @4" for H/6 or atleast lateral dimension or 1'6" whichever is greater & 8ø @6"c/c at remaining height	####	
Third FL			400	400	1666		8	4	2061	1.24		1.29	
Second FL			400	400	1346		8	4	2061	1.53		1.29	
First FL			400	400	1332		12		2413	1.81		1.51	
Ground Floor			400	400	2367	4	8		2865	1.21		1.79	

SPECIAL CONFINING REINFORCEMENT

The amount of special confining reinforcement required in columns shall be calculated as follows:

L_0 = *maximum* (Dc; Lc/6 ; 450mm).....4.3.1

Where,

D_c = Largest cross sectional dimension of column= 350mm

l_{cl} = Clear length of column= 2843.645 lc/6= 473.9409

Hence, Provide Special confining bar up to 480mm on either side of column

The spacing of the confining reinforcement shall be as follows:

Spacing =Minimum(1/4 Bc; 6dc; 100mm)

Where,

B_c = Smallest cross sectional dimension of column= 350mm

d_b = Diameter of smallest longitudinal bar of column= 16

6dc=

Hence, Provide spacing of 100mm

in rectangular links:

A_{sh} = Maximum of {0.18Sv h fck/fy*(Ag/Acc-1) = 78

0.05 Sv h fck/fy = 60

here,

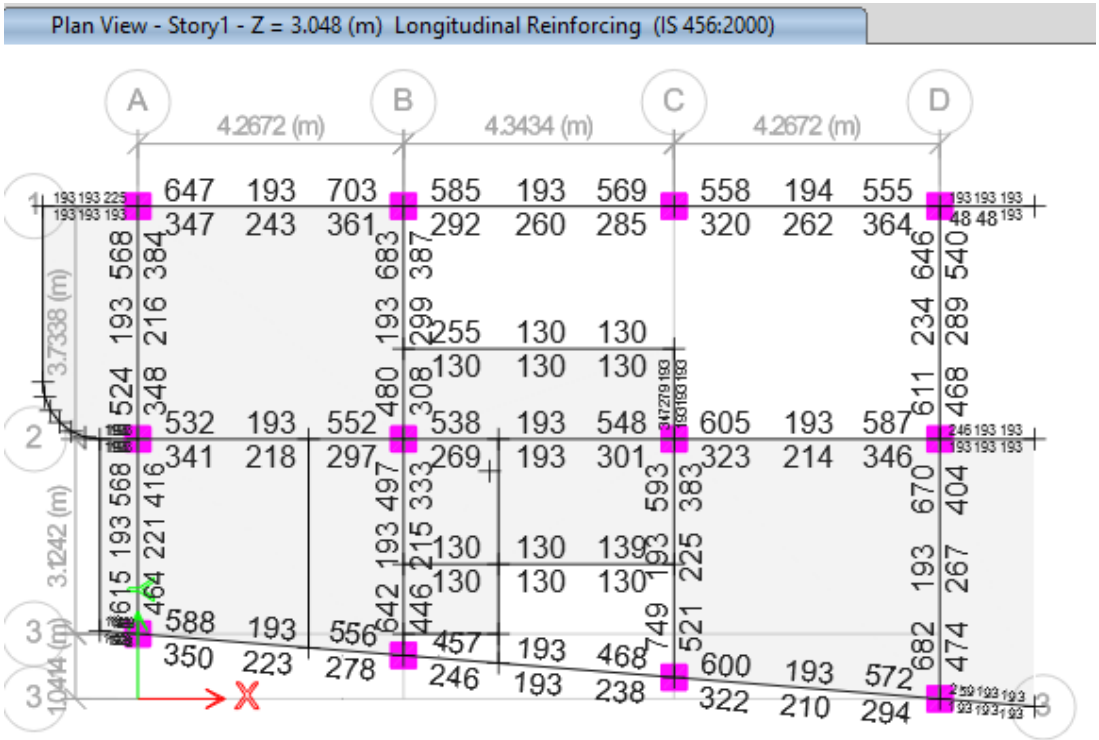
h = longer dimension of rectangular link/hoop measured to its outer face, which does not exceed 300 mm

A_{cc} = area of confined concrete core in rectangular link/hoop measured to its outer dimensions. A_{cc} shall be measured as overall core area, regardless of link/hoop arrangement.

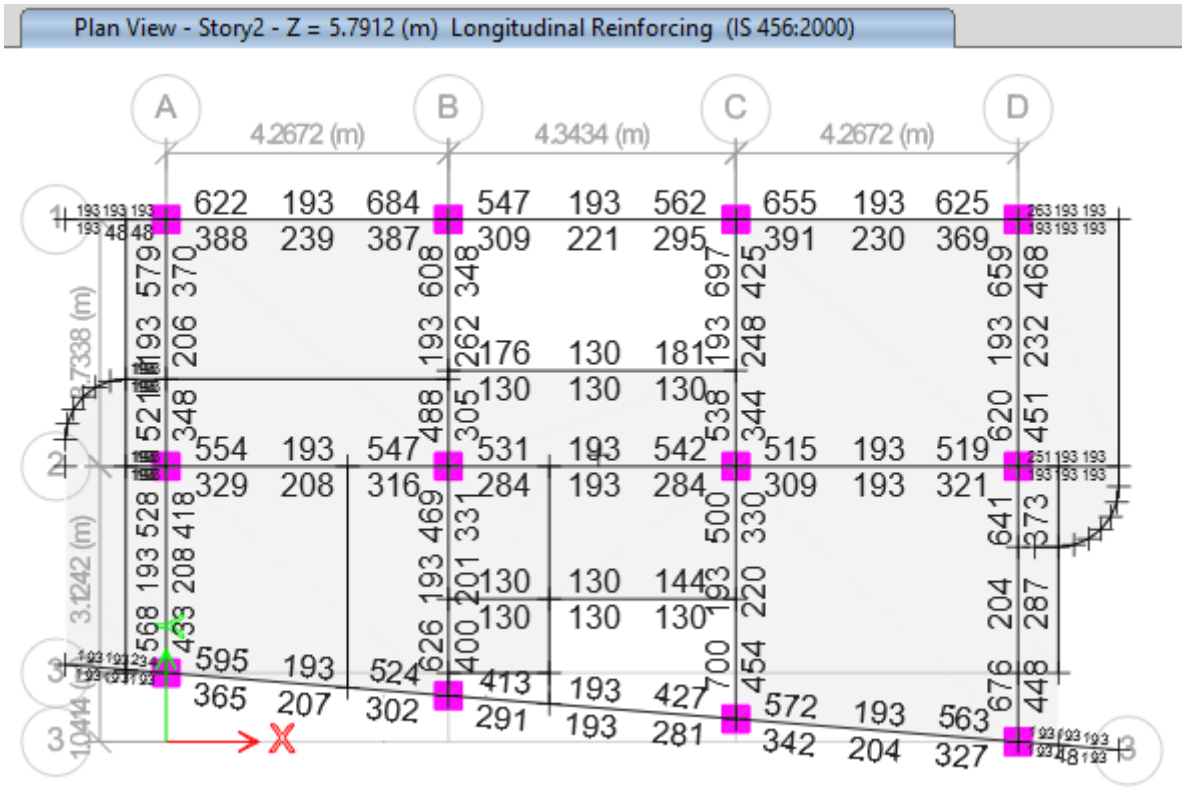
Hence, provide bar dai of 8mm bar

Colm Shear Reinforcement						
Design for shear reinforcement						
shear reinforcement req Ast				=	394.16	mm^2/m
depth of beam				=	350	mm
width of beam				=	350	mm
Dia of stirrups				=	8	mm
no of leg stirrup provided =				=	2	nos
Area of leg stirrup provided = ast				=	100.531	mm^2
Spacing req = 1000*ast/Ast				=	255.0512	mm^2
Spacing provided				=	150	mm
Provided Shear Reinforcement				=	1005.31	mm^2

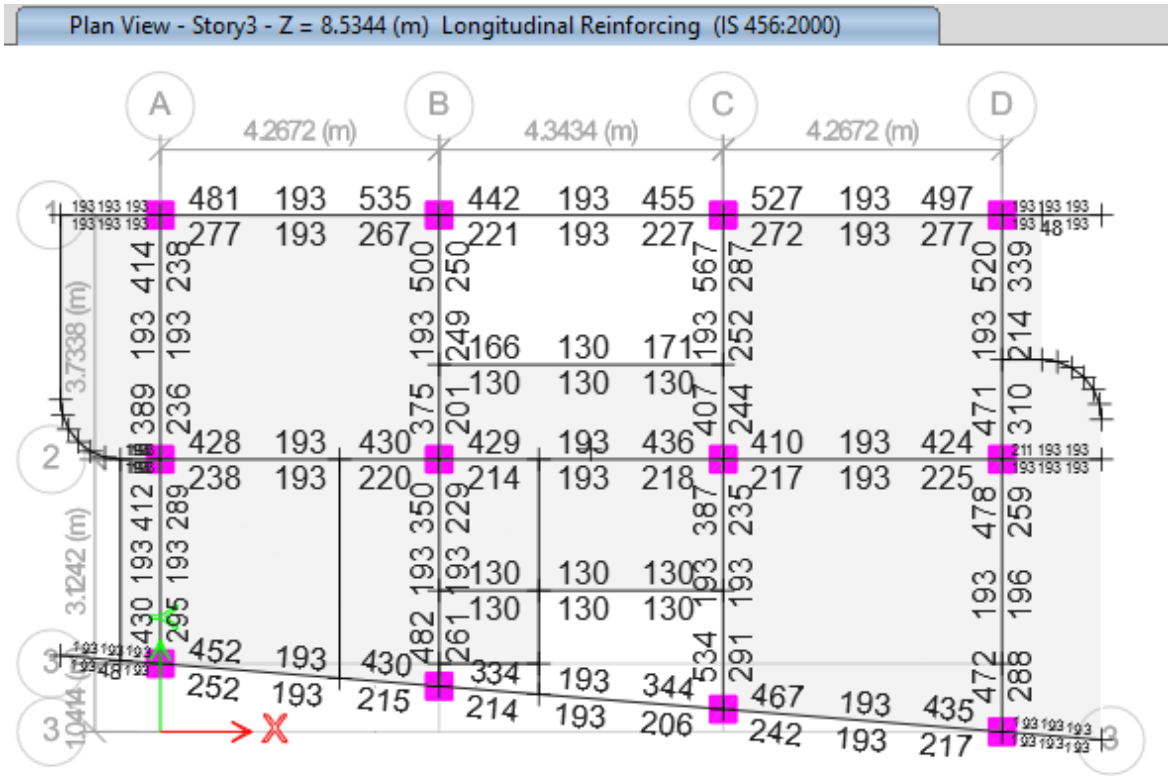
Beam Design Summary



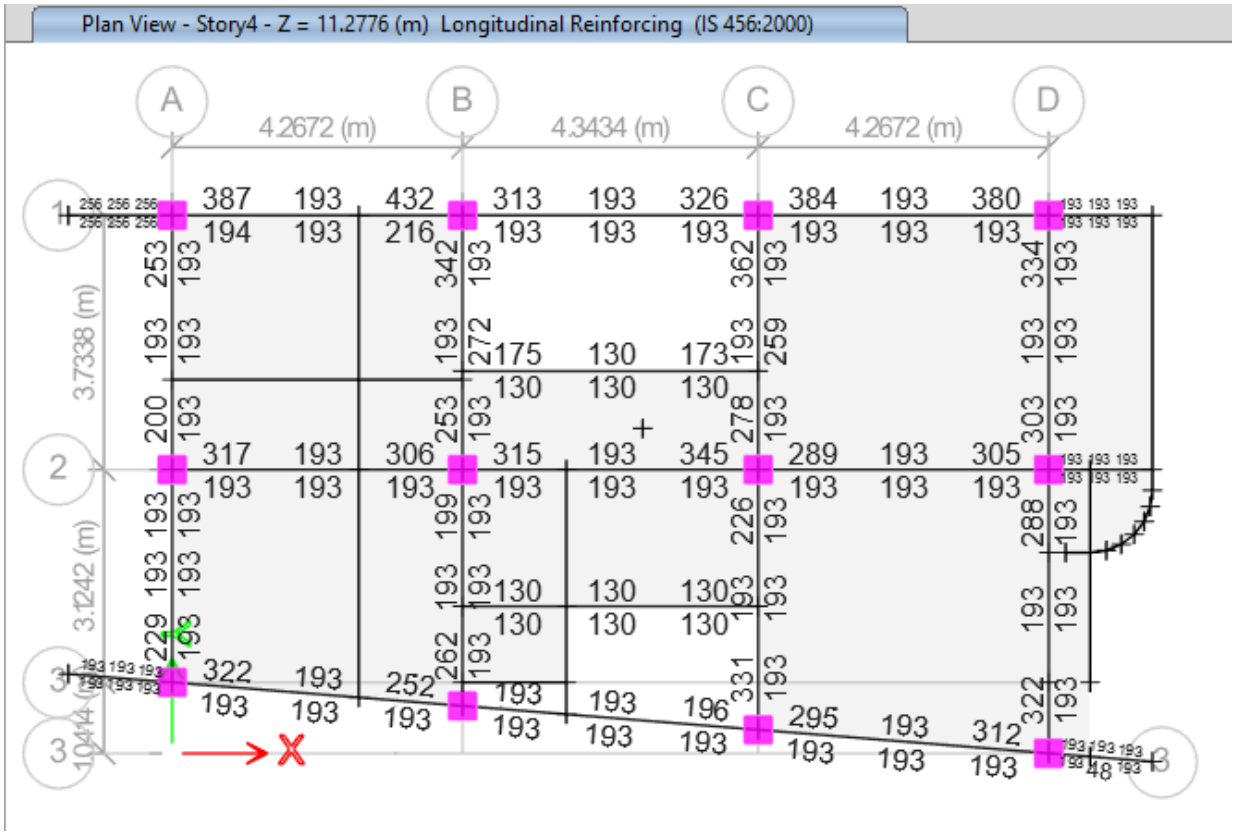
At First floor Level



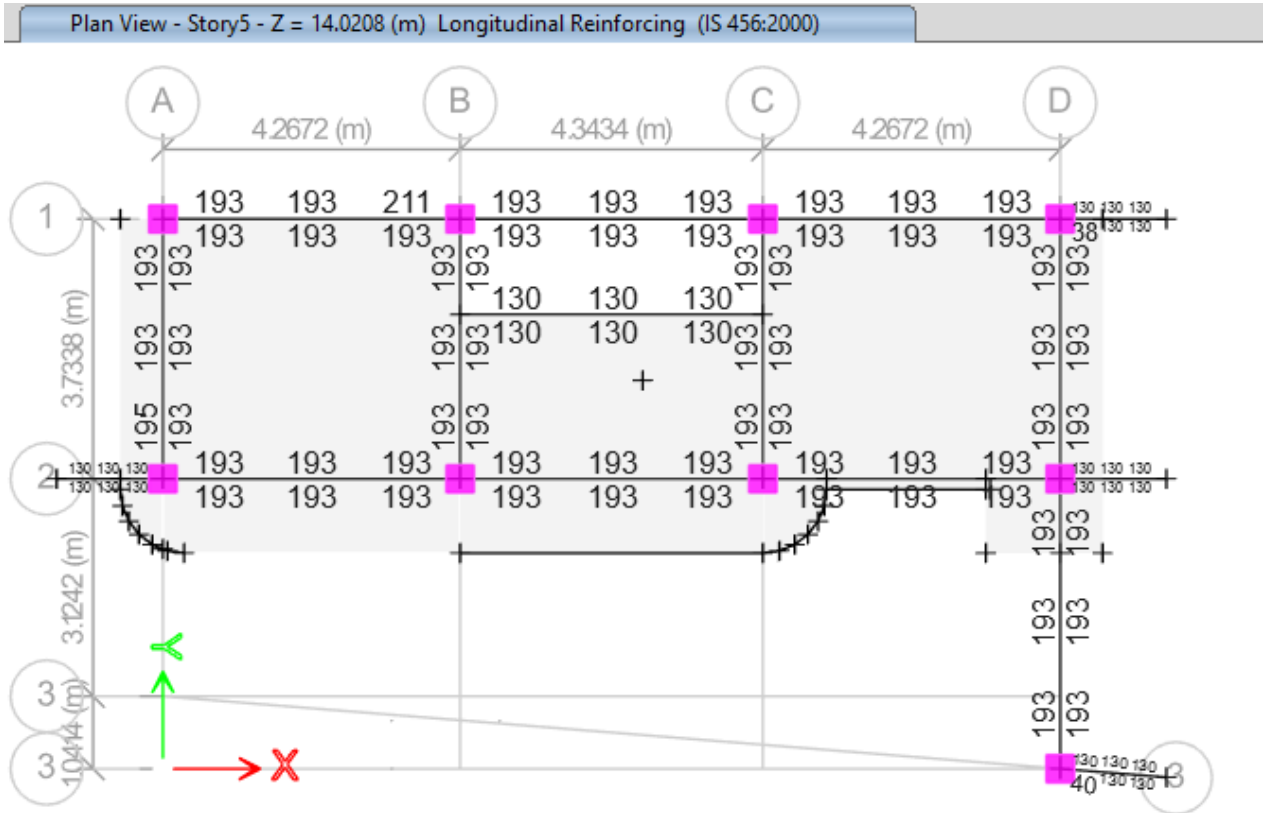
At Second floor level



At Third floor Level



At Fourth floor Level



At Top floor Level

Table 0-2: Beam Design Summary

Primary Beam Design

Concrete Grade M20

Steel Grade Fe500

Storey Lev.	Beam	Bar location	Width b	Depth D	Bar position	Area from Etabs mm²	No of 16ø		No of 12ø		Tot Ast prov mm²	Stirrups	Remark
							Th	ex	Th	ex			
Critical Beam along X-X direction Grid 1-1													
At 1st Floor	A1-B1	Support	9"	17"	top	622	2		1	2	741.4	10ø stirrups @4"c/c at support & 10ø stirrups @6" c/c at mid section	
					bot	388	2		1		515.2		
		Mid			top	193	2		1		515.2		
					bot	239	2		1		515.2		
		Support			top	684	2		1	2	741.4		
					bot	387	2		1		515.2		
	B1-C1	Support	9"	17"	top	547	2		1	2	741.4	10ø stirrups @4"c/c at support & 10ø stirrups @6" c/c at mid section	
					bot	309	2		1		515.2		
		Mid			top	193	2		1		515.2		
					bot	221	2		1		515.2		
		Support			top	562	2		1	2	741.4		
					bot	295	2		1		515.2		
	C1-D1	Support	9"	17"	top	655	2		1	2	741.4	10ø stirrups @4"c/c at support & 10ø stirrups @6" c/c at mid section	
					bot	391	2		1		515.2		
		Mid			top	193	2		1		515.2		
					bot	230	2		1		515.2		
		Support			top	625	2		1	2	741.4		
					bot	369	2		1		515.2		
Critical Beam along Y-Y direction Grid C-C													
At 1st Floor	C1-C2	Support	9"	17"	top	697	2		1	2	741.4	10ø stirrups @4"c/c at support & 10ø stirrups @6" c/c at mid section	
					bot	425	2		1		515.2		
		Mid			top	193	2		1		515.2		
					bot	248	2		1		515.2		
		Support			top	538	2		1	2	741.4		
					bot	344	2		1		515.2		
	C2-C3	Support	9"	17"	top	500	2		1	2	741.4	10ø stirrups @4"c/c at support & 10ø stirrups @6" c/c at mid section	
					bot	330	2		1		515.2		
		Mid			top	193	2		1		515.2		
					bot	220	2		1		515.2		
		Support			top	700	2		1	2	741.4		
					bot	454	2		1		515.2		

Page 1

For more details refer structural drawing.

Design for shear reinforcement

shear reinforcement req Ast	=	1275	mm^2/m

depth of beam	=	375	mm
width of beam	=	230	mm
Dia of stirrups	=	8	mm
no of leg stirrup provided =	=	0	nos
Dia of stirrups	=	10	mm
no of leg stirrup provided =	=	2	nos
		157.079	
Area of leg stirrup provided = ast	=	6	mm^2
		123.199	
Spacing req = 1000*ast/Ast	=	7	mm^2
Spacing provided	=	100	mm
		1570.79	
Provided Shear Reinforcement	=	6	mm^2

Hence, 2L-10dia stirruos are provided.

Slab Design Summary

Depth of Slab: 5”
Reinforcement provide 8mm@6" c/c both ways with usual layouts/curtailments wherever required
For more details refer structural drawings.

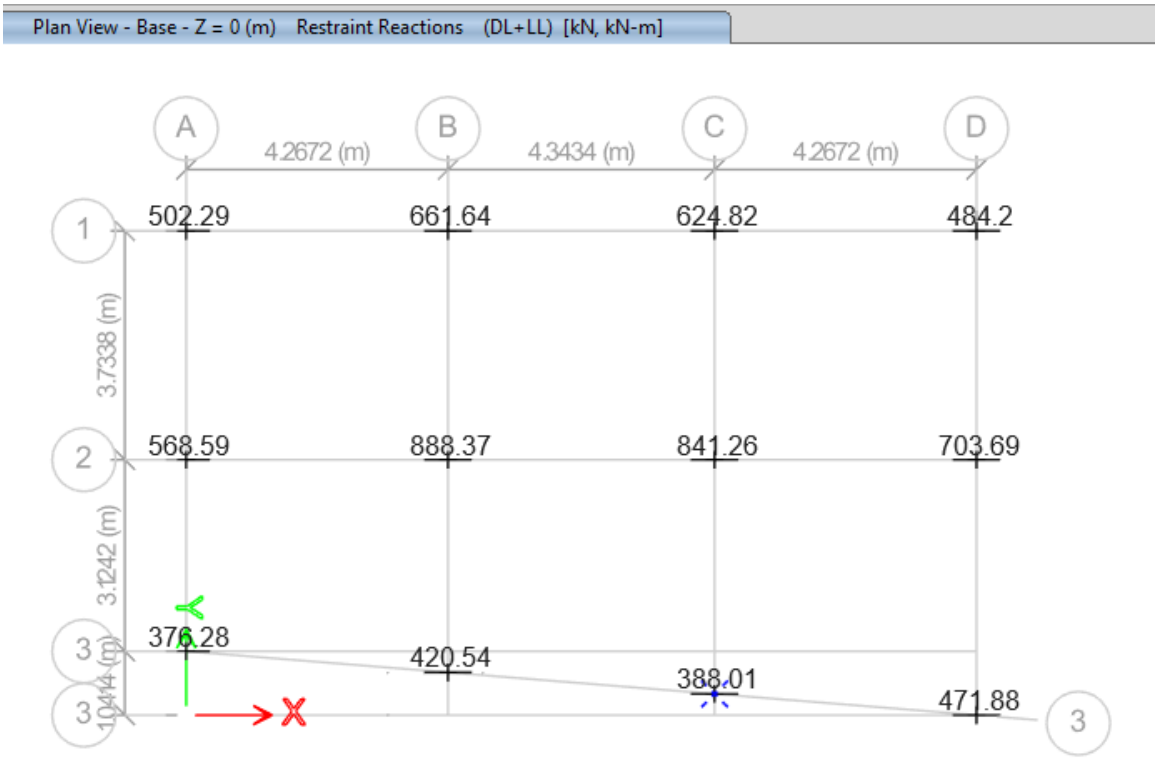
Staircase Design Summary

Waist Slab Thickness: 5" Overall
Clear cover: 20 mm
Longitudinal reinforcement: 12 mm bars@ 6" c/c
Transverse Reinforcement: 8 mm bars @ 6"c/c
Size of landing beam:9" x 6"
Reinforcement in landing beam: 2-12 ϕ at top and bottom
With 8ϕ 2-legged stirrups @ 4" c/c
For more details refer structural drawing.

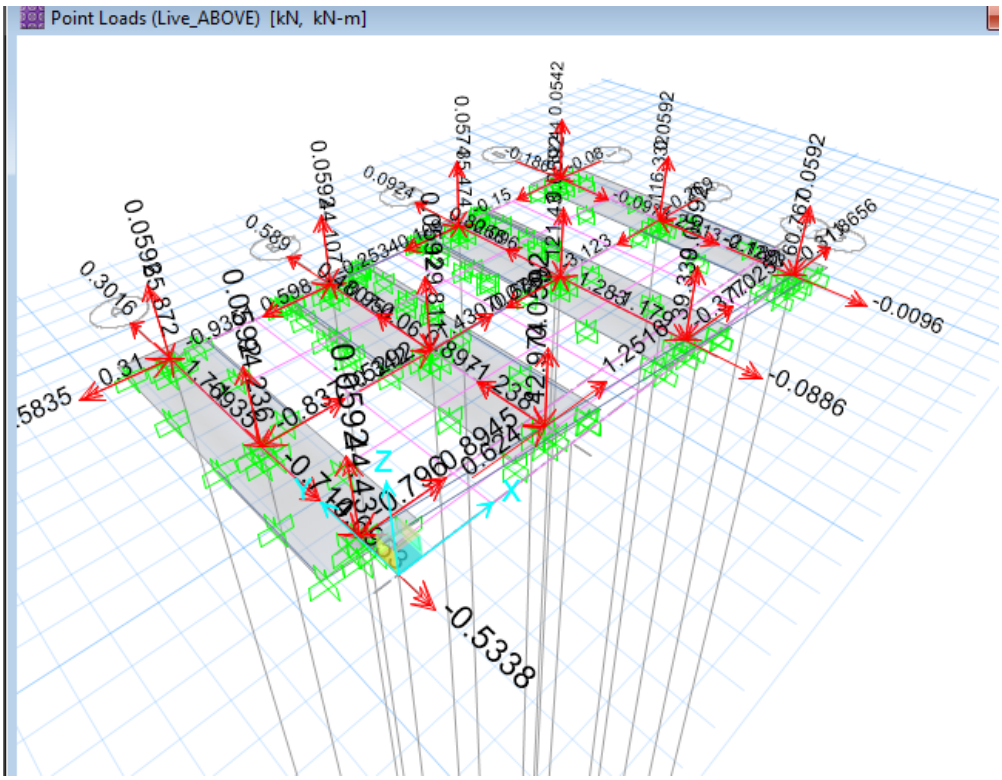
Foundation Design Summary

The foundation is designed as MAT Footing. The footing base is to be formed at 5’6” ' below ground level.

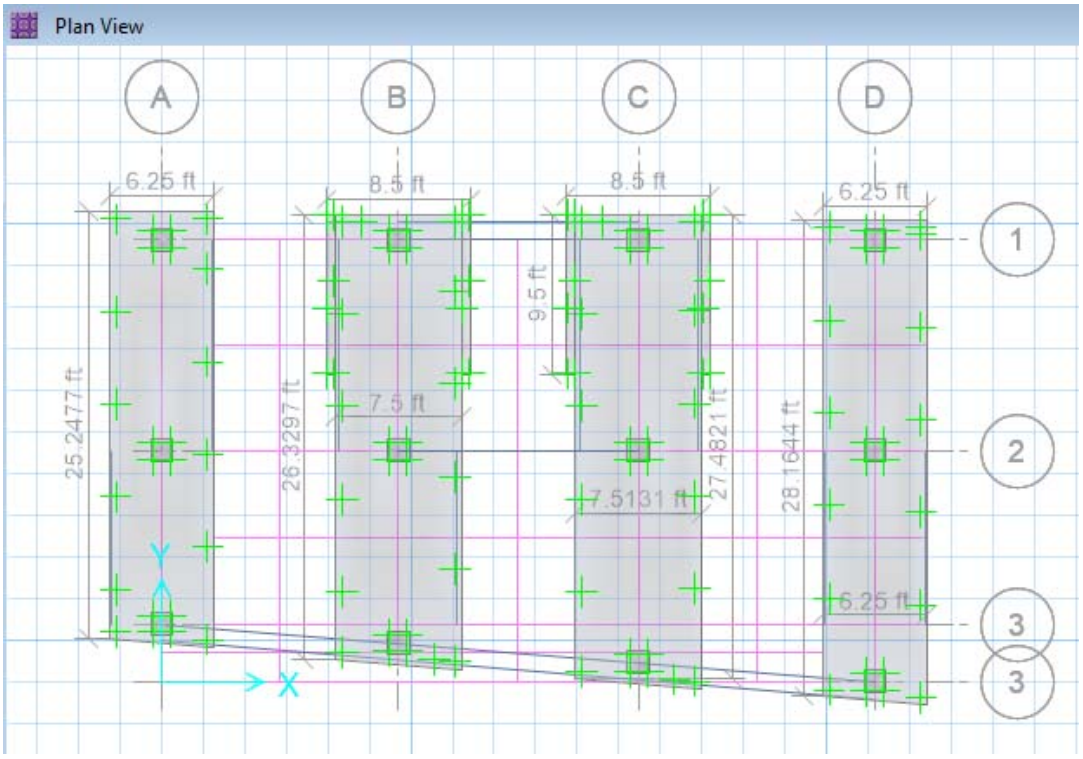
The footing is modeled as a thick plate. The position of columns is located and loads from superstructure are applied to it.
Soil Bearing Capacity (SBC) =120 kN/m^2
The modulus of subgrade is computed using Bowles equation as follows
Modulus of Subgrade = 40xSBCxSafety Factor= 40x130x3 = 15600 kN/m^3.
Concrete Grade= M20 (f_{ck}= 20 Mpa) for Mat
Reinforcement Steel Grade = Fe500 (f_y = 500 Mpa



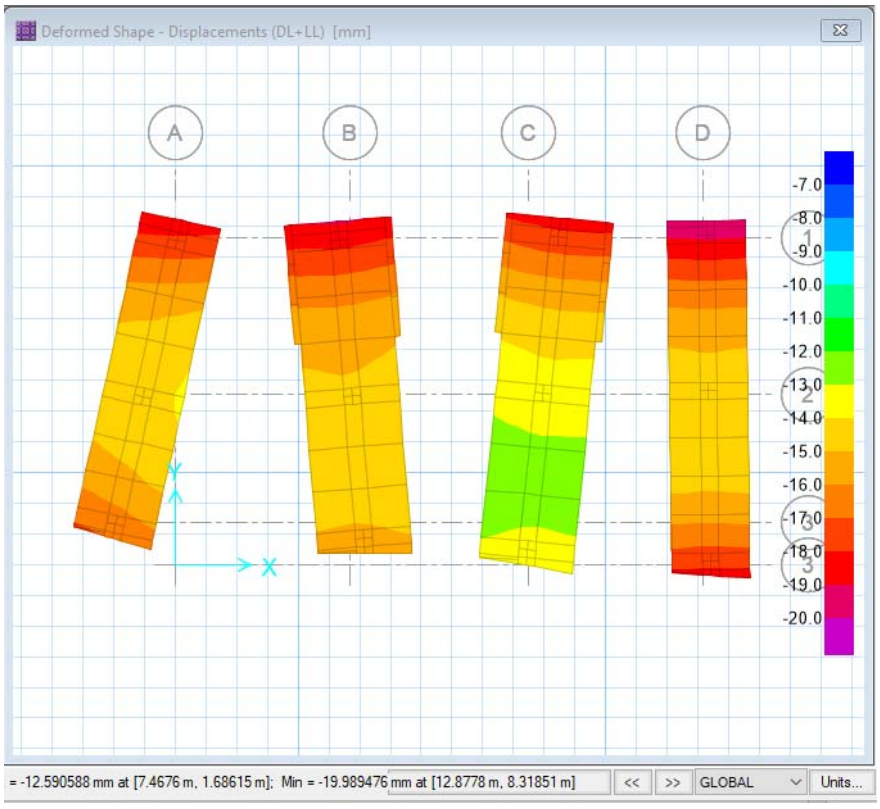
Base Reaction At Dl +LL



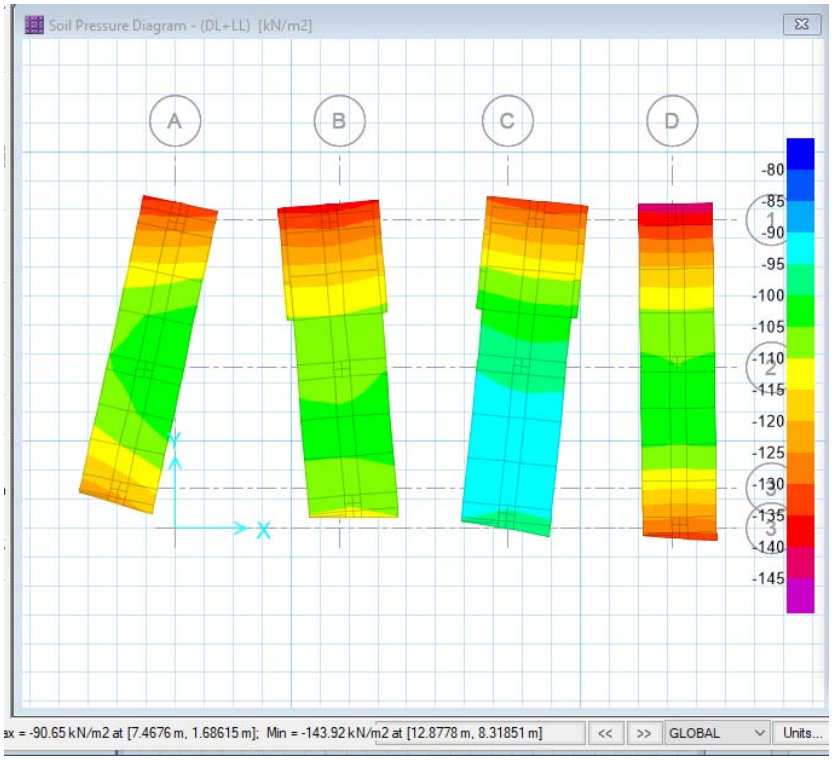
Footing Model in Safe



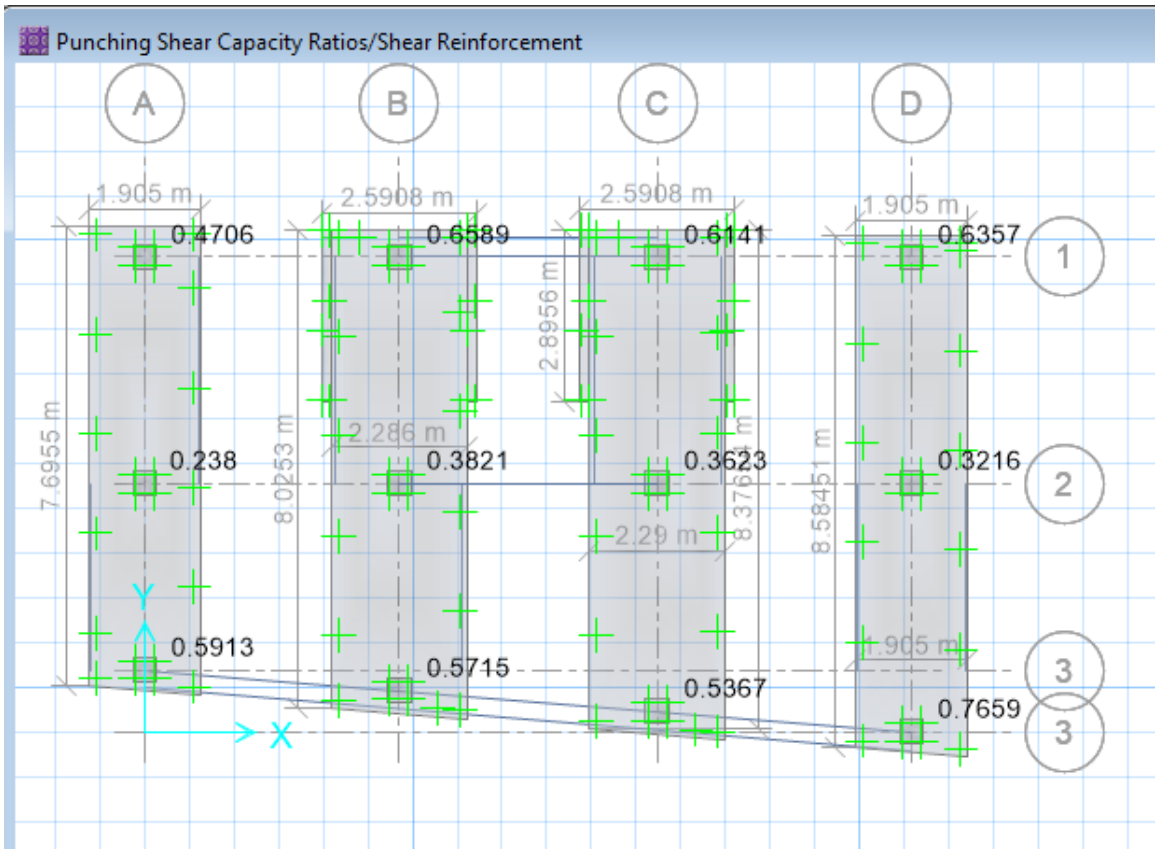
Footing Model in Safe



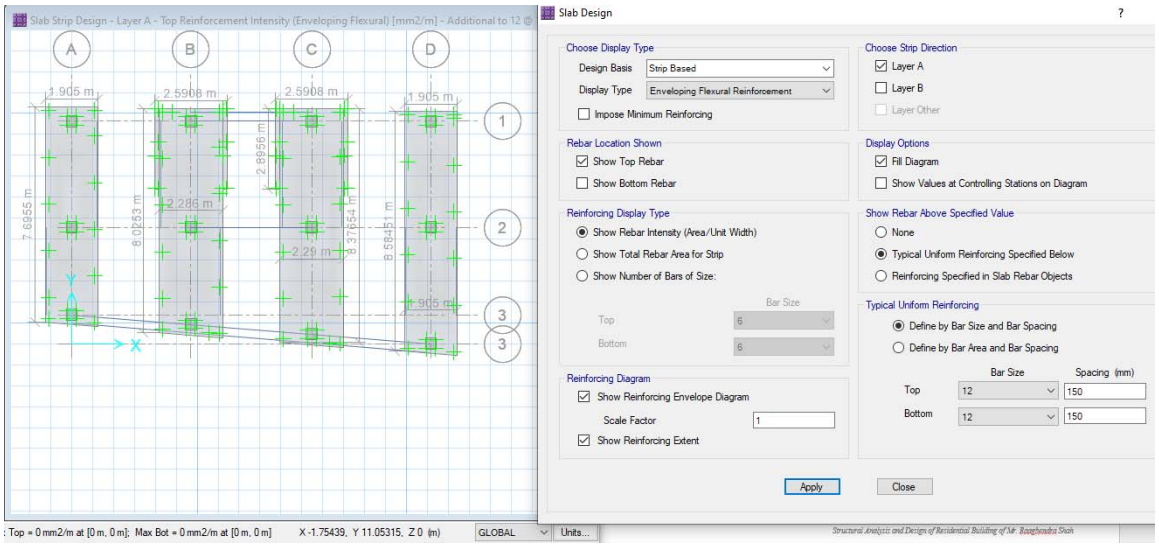
Footing displacement at DL+LL

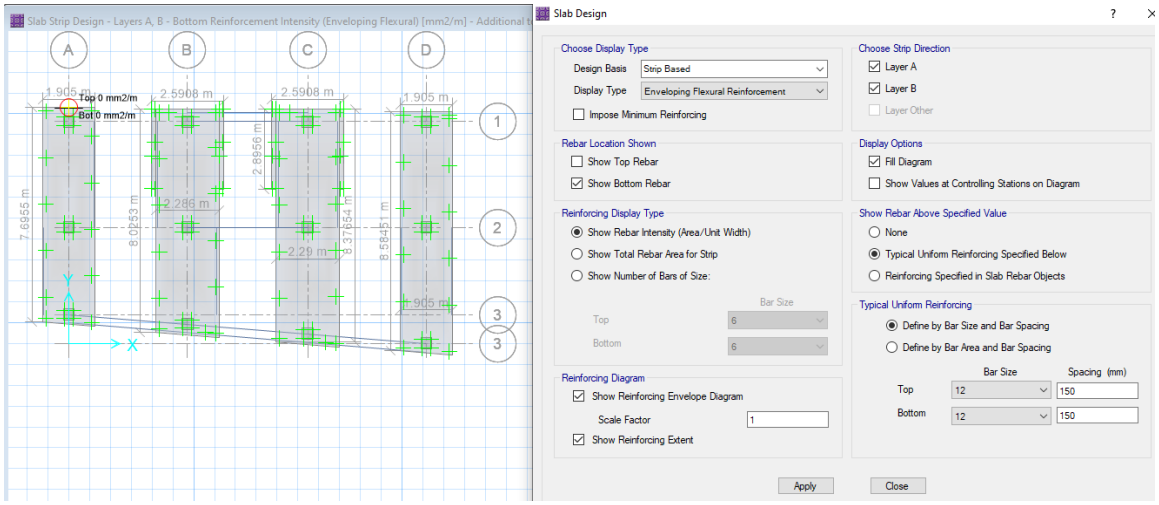


Soil Pressure at DL+LL



Punching Shear

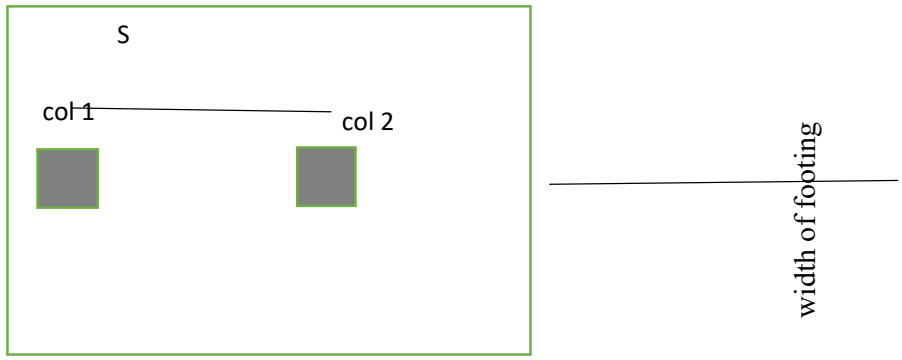


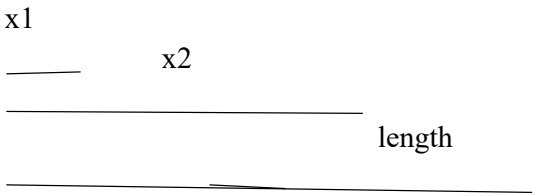


Slab Design of Footing

Combine Footing Design CF1
At C2-C3

Bearing Capacity =	120	KNm						
Fy=	500							
Fck =	20							
dimension	b(m)	l(m)						
col1 =	0.4	0.4						
col2 =	0.4	0.4						
1) Length of footing								
UnFactored Load from	**col1**	**col2**						
	635.0	*835*	KN					
s' =	0							
x1 =	*0.2*							
S =	3.81							
x2 =	*4.01*							
X =	2.36							
Length =	4.73		Adopt	***4.9***	m		15.6	
Total Load =	1470	KN		3				
Wt. of footing including Earth		=	1617	KN				
width of footing =	2.44		Adopt	***2.5***	m		9.51	2.831858
2) Longitudingal Bending moment and shear								
Factored Load from		**col1**	**col2**					
		952.5	*1253*	KNm				





net upward soil pressure=	106.715		
	ok		

soil pressure for shear		=	**160.1**	KN/m^2		
soil pressure for shear per unit length		=	464.2	KN/m		
max shear force at the centre line of col 1						
		v1 =	**-92.8**	KN		
		v2 =	**859.7**	KN		
max shear force at the centre line of col 2						
		v1 =	**343.5**	KN		
		v2 =	**-909**	KN		
Point of Zero shear force from centre of the col 1						
		x =	**1.85**	m		
Max BM computed from left side		=	-787	KNm		
Max BM computed from Right side =			-763	KNm		
1.96	0.74					
Design BM =	**786.7**	KNm				
depth from BM consideration =			**319.3**	mm		
			0.32	m		
Adopt D =	500	mm	cover =	50	mm	
Adopt d =	450	mm				
	0.45	m				
		B =	2900	mm		
		D =	500	mm		
		d =	450	mm		
For BM =	786.70	KNm				
Required Ast =	4387.7	mm2				
Min ast =	0.12	%	1566	mm2		
Final Required ast =		4388	mm2			
Area of bar dia =	16	=	201.1			

A	**3.75**
B	**-2.0E+05**
C	**7.87E+08**
Ast =	**4387.736**

					Ideal Ast provided		
	No of bars required =		21.82	22	=	4E+03	mm2

This Ast is true When the Bar 16 is provided at spacing 133.333333333333 mm

cover =	50		
spacing =	133.3	mm	
Adopted spacing of the bars for the longitudinal bar at the top =		150	mm
		6	inch

no of Bars=	**29.5**
Ast Pro=	**5931**

i) shear one way action

Max shear force occur in		Col 2			
col size (b)=	0.4	m	col size (l) =	0.4	
Design SF(KN) =	908.98				
Vu =	607.25	KN			
		Vu/bd			
Nominal Shear Stress	Tv =	=	0.46	N/mm2	
P% =	0.45	%			

	X	Y
	% Ast	Tauc
X1,Y1	0.5	0.48
X,Y	0.45	***0.46544***
X2.Y2	0.75	0.56

Table 19

Shear Strenght for the M20 concrete = 0.465 N/mm2

OK

ii) shear two way

shear strenght of concrete		Tc' =Ks*Tc			
betta =	1		short =	0.4	
trial Ks =	1.5		Long =	0.4	
Ks =	1				
concrete grade =	M	20			
Tc =	1.118				
Tc' =	1.118				
shear force Vu =	1190				

Nominal shear stress	Tv =	Vu/bo/d =		0.78		
OK						
3)	*Positive longitudinal reinforcement at the bottom of footing beyound column faces*					
	BM at face of col 1 =		0.00	KNm		
	BM at face of col 2 =		67.7	KNm	0.54	
	M20	Fe 500	B =	2900	mm	
			D =	500	mm	
			d =	450	mm	
	Design BM =	**67.68**	KNm			
	Required Ast =	348.1	mm2			
	Minimum Ast =	0.12	%	1566	mm2	
	Final Required Ast =		**1566**	mm2		
	Area of bar dia =	12	=	113.1		
	No of bars required =		13.85	14		
	cover =	50				
	spacing =	215	mm			
	Adopted spacing of the bars for the longitudinal bar at the bottom =				150	mm
					6	inch

A	**3.75**
B	**-195750**
C	**67681895**
Ast =	**348.0778**

4)	*Transverse Reinforcement*				
FOR	***COL 1***				
	col size (m) =	0.35			
	d (footing) (m) =	0.45			
	Band width =	1.25	m		
	Bar dia provided in longitudinal dir =			12	mm
	Effective d in transverse dir		0.438	m	

=

B = **1250** **mm**

d = **438** **mm**

A	**8.7**
B	**-190530**
C	**2.67E+08**
Ast =	**1504.543**

Factored upward pressure under col 1 = 328.4 KN/m

BM at the face of col 1 = **267.0** KNm

1.275

Ast = 1504.5 mm2 for BM = 267.0 KNm

Area of bar dia = 12 = 113.1

No of bars required = 13.30 14

cover = 50

spacing = 88 mm

Adopted spacing of the bars for the transverse bar at the bottom = 150 mm

6 inch

FOR

COL 2

col size (m) = 0.35

d (footing) (m) = 0.45

Band width = 1.25 m

Bar provided in longitudinal dir = 12 mm

Effective d in transverse dir

= 0.438 m

B = **1250** **mm**

d = **438** **mm**

A	**8.7**

Factored upward pressure under col 2 = 431.9 KN/m

B	**-190530**
C	**3.51E+08**
Ast =	**2030.817**

BM at the face of col 1 = **351.1** KNm

1.275

Ast = 2030.8 mm2 for BM = 351.1 KNm

Area of bar dia = 16 = 201.1

No of bars required = 10.10 10.5

cover = 50

spacing = **121** mm

Adopted spacing of the bars for the transverse bar at the bottom = 150 mm

TRANSVERSE AT THE TOP

6 inch

Minimum Ast = 0.12%

657

Area of bar dia = 12 = 113.1

No of bars required = 5.81 6

cover = 50

spacing = **230** mm

Adopted spacing of the bars for the transverse bar at the top = 150 mm

6 inch

5 CONCLUSION

After the analysis of the building components, the building is found to be safe against the gravity as well as Seismic Loads. The section sizes and reinforcements are sufficient enough to withstand all kinds of possible axial, shear, flexural and torsional forces. The design is good enough to assure them Life Safety level of performance under the Design Basis Earthquake considered for Zone factor 0.35, Importance Factor 1. Bearing capacity of the soil is 120 kN/m^2. M20 grade of concrete is used for columns, beams, & M20 for intermediate beam, slabs and foundations.

6 RECOMMENDATIONS

Design and construction of the structure are inter–related jobs. A performance of a building depends upon a work and material quality during the construction more than the intensions pursued during structural design. A large percentage of structural failures are attributed due to poor quality of construction. Therefore, to assure the proper safety, material and work quality should be maintained during the construction. ***Structural designer will be responsible for all the designs but not for any faulty constructions happened at site beyond his supervision***. Following recommendations are made by the structural designer.

1. It is recommended to strictly follow the section sizes and reinforcements provided in the structural drawings.
2. It is recommended that the site engineer should be responsible to handle the problems that may arise during construction. He/she shall also be responsible for maintaining the material and process quality during construction.
3. It is strictly recommended that any changes in the design shall be done only with the consultation of the structural designer.
4. It is strictly recommended to maintain the standards in the grade of cement and reinforcement steel. It is recommended to run a cube test to ensure the strength and quality of the concrete ratio used. It is also recommended test the reinforcement steels to ensure the quality of steel used.
5. The building foundation is designed assuming the Soil Bearing Capacity to be 120 KN/m^2. However, to ensure that the assumed value is safe it is recommended to verify the assumed value with suitable methods.

ANNEXES

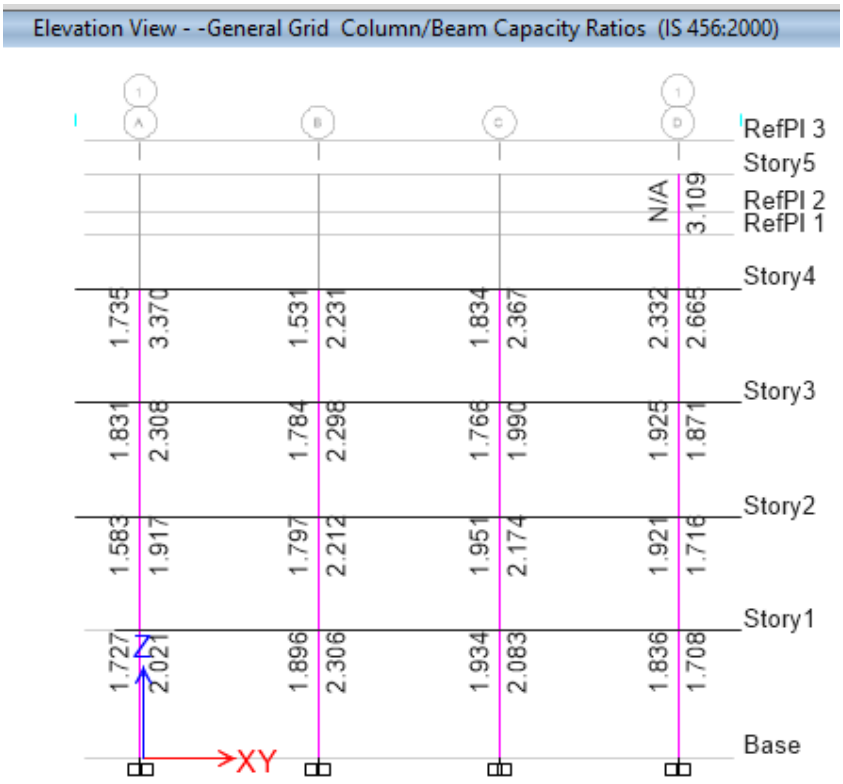
ANNEX -I: REFERENCES

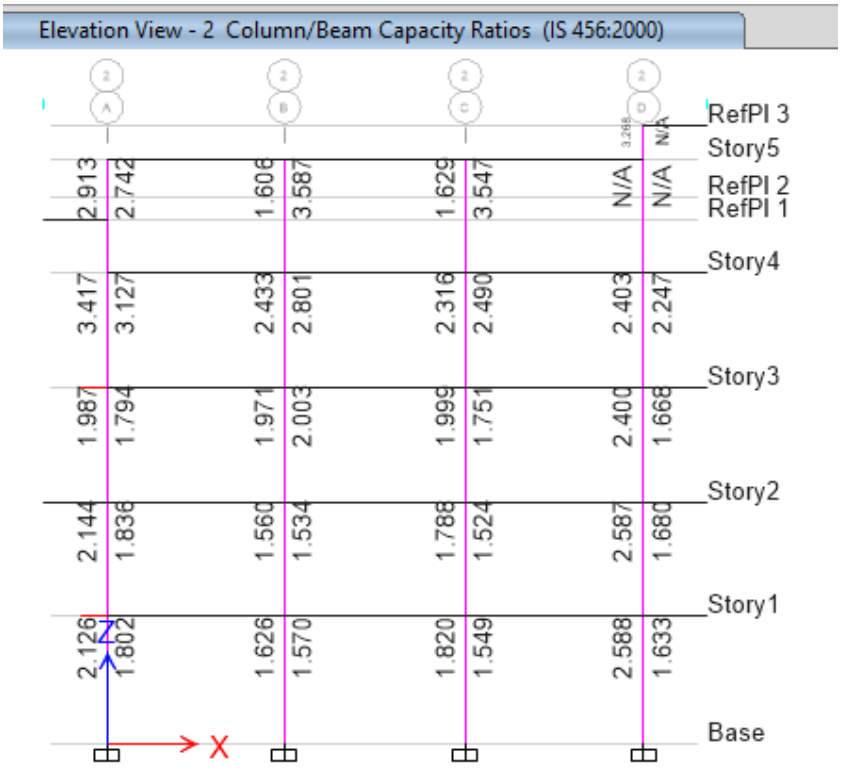
IS: 456 – 2000	Code of Practice for Plain and Reinforced Concrete
IS: 875 (Parts 1-5)	Code of practice for design loads (other than earthquake) for buildings and structures (second revision) Part 1 – Dead loads

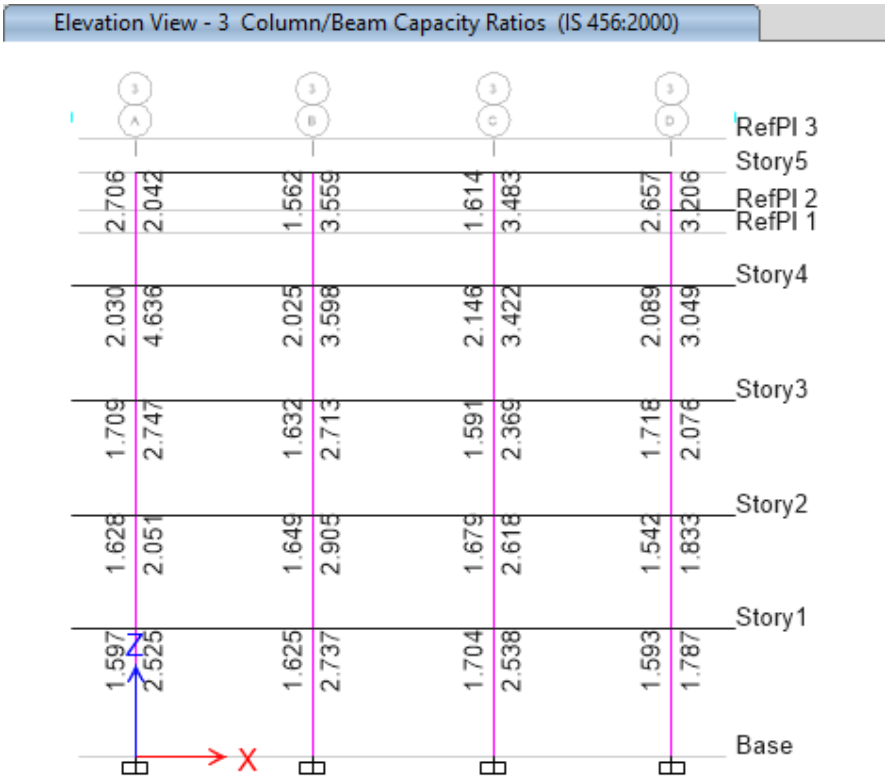
	Part 2 – Imposed loads
NBC 105: 2025	Nepal National Building Code
IS: 13920 - 2016	Ductile Detailing of Reinforced Concrete Structures subjected to Seismic forces - Code of Practice
IS: 4326-1993	Code of Practice for Earthquake Resistant Design of Structures
SP: 16 – 1980	Design Aids for Reinforced Concrete to IS: 456 – 1978
SP: 34 – 1987	Handbook on Concrete Reinforcement Detailing
Jain, A.K.	Reinforced Concrete, Limit State Design, fifth edition, Nem Chand and Bros, Rookie, 1999
Sinha, S. N.	Reinforced Concrete Design, Second edition, Tata McGraw Hill Publishing Company Ltd, New Delhi, 1996
	Reinforced Concrete Design, Second edition, Tata McGraw Hill Publishing Company Ltd, New Delhi, 2003
Pillai,U.C. & Menon,D.	Advanced Foundation Engineering Design Equations for Strap Footings
Dr. Adnan, A. Basma	

7 Column to beam Capacity Ratio For Different Beam Column Joint

Column to Beam Capacity Ratio for different B/C joints as obtained from the check performed in Etabs 2018.0.2 is presented as follows.
Note: B/C ratio check for a joint is not significant when there is no structure above the joint. The B/C joints in which which (Column to Beam Capacity Ratio) *less than the limiting value 1.2. at the top floor are hence not significant.*


Elevation View - -General Grid Column/Beam Capacity Ratios (IS 456:2000)
RefPl 3
Story5
RefPl 2
RefPl 1
Story4
Story3
Story2
Story1
Base
N/A
3.109
1.735
3.370
1.531
2.231
1.834
2.367
2.332
2.665
1.831
2.308
1.784
2.298
1.766
1.990
1.925
1.871
1.583
1.917
1.797
2.212
1.951
2.174
1.921
1.716
1.727
2.021
1.896
2.306
1.934
2.083
1.836
1.708
Z
XY


Elevation View - 2 Column/Beam Capacity Ratios (IS 456:2000)
RefPl 3
Story5
RefPl 2
RefPl 1
Story4
Story3
Story2
Story1
Base
N/A
N/A
2.913
2.742
1.606
3.587
1.629
3.547
3.417
3.127
2.433
2.801
2.316
2.490
2.403
2.247
1.987
1.794
1.971
2.003
1.999
1.751
2.400
1.668
2.144
1.836
1.560
1.534
1.788
1.524
2.587
1.680
2.126
1.802
1.626
1.570
1.820
1.549
2.588
1.633
Z
X


Elevation View - 3 Column/Beam Capacity Ratios (IS 456:2000)
RefPl 3
Story5
RefPl 2
RefPl 1
Story4
Story3
Story2
Story1
Base
2.706
2.042
1.562
3.559
1.614
3.483
2.657
3.206
2.030
4.636
2.025
3.598
2.146
3.422
2.089
3.049
1.709
2.747
1.632
2.713
1.591
2.369
1.718
2.076
1.628
2.051
1.649
2.905
1.679
2.618
1.542
1.833
1.597
2.525
1.625
2.737
1.704
2.538
1.593
1.787
Z
X

Sample Design of Slab

Design Data

f_{ck} = 20 N/mm^2

f_y = 500 N/mm^2

Dimensions of the slab (c/c distance bet^n supports),

Length of short span, l_x = 3.810 m

Length of long span, l_y = 4.260 m

Width of the supporting beam, = 230 mm

Clear cover to main reinforcement = 15 mm

Assume dia. of reinforcement steel = 8 mm

30.48

Calculations

Assume the thickness of slab as **125** m
m ;

Effective depth, d_{eff} = 106 m
m

Effective span, l_x = 3.81 m (or) 3.686 m whichever is less; = **3.686** m

l_y = 4.26 m (or) 4.136 m whichever is less; = **4.136** m

(l_y / l_x) = $\frac{1.1}{2}$ < 2 ; Here, (l_y / l_x) is less than 2

Hence design the slab as two way slab

Load Calculations

Dead Load of slab = 0.125 x 25 = 3.13 KN/m^2

Finishes load on slab = 1.25 KN/m^2

Total Dead load acting on the slab = 4.38 KN/m^2

Live Load on slab = 2.0 KN/m^2

Factored Design Load, W = **9.57** KN/m^2

Support Condition (Type of panel according to support condition)

Two Short Edges Discontinuous — For this support condition,

Short span coefficient for (l_y / l_x) = 1.12, Long span coefficient,

For negative moment, a_x = 0.050 | For negative moment, a_y = 0

For positive moment, a_x = 0.038 | For positive moment, a_y = 0.035

Moment Calculation

Max. BM per unit width, $M_x = a_x w l_x^2$ & $M_y = a_y w l_x^2$

	M_u KNm	M_u / bd^2 N/mm^2	p_t %	$A_{st,\ req}$ mm^2
For Short Span,				
At mid span,	4.89	0.44	0.1039	110
At supports,	6.45	0.57	0.1357	144
For Long span,				
At mid span,	4.55	0.47	0.1112	118
At supports,	4.89	0.51	0.121	128

$A_{st,\ min}$ = (0.12/100) bD = 150 mm^2

Reinforcement details

Provide **8** @ **150** mm c/c at midspan & supports for short span (A_{st} pro. = 335 mm^2)

Provide **8** @ **150** mm c/c at midspan & supports for long span (A_{st} pro. = 335 mm^2)

Check for Deflection

Percentage of tension reinforcement = 0.27 %

f_s = $0.58\ f_y\ (A_{st\ req} / A_{st\ pro})$ = 95

Refer Fig. 4 of IS 456,

Modification factor = 1.1

$(l/d)_{max} = (l/d)_{basic} \times k_t \times k_c$ = 28.6

$(l/d)_{provided}$ = 35

< (l/ d) max (i.e. = 28.6) **OK**

Hence safe in deflection.

Check for shear

Shear force per *m* strip ($W*l_x/2$), V = 17.638 KN/ m

Shear stress , (V/bd), τ_v = 0.140 N/mm^2

P_t = 0.27%

Design Shear Strength of Concrete

τ_c = 0.360 N/mm^2 Refer Table 19, IS 456

> τv (i.e. = 0.14) **OK**

Since $\tau_c > \tau_v$, so safe in shear.

SAMPLE STAIRCASE DESIGN

Design of Staircase

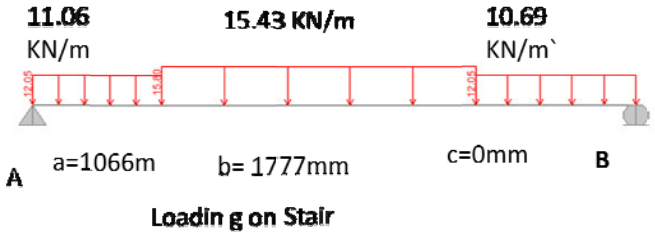
Grade of concrete fck =	**20**	**N/mm²**	**Grade of Steel fy=**	**500**	**N/mm²**
Unit Wt. of concrete γc=	**25**	**KN/m³**			
Unit Wt. of brick γm=	**19.2**	**KN/m³**			

Live Load LL = **3** **KN/m²**
Floor Finish FL = **1.25** **KN/m²**

Reference	Step	Calculation	Output	
	1	**Known Data**		
		Total height between floor H =	2.868812	m
		Width of Stair B =	1.143	m
		Width of Landing W =	1.143	m
		Riser R =	177.8	mm
		Tread T =	254	mm
		c/c length of stair L =	2.844	m
		bearing width w1 =	300	mm
		bearing width w2 =	300	mm
		Stair section a =	1.066813	m
		Stair section b =	1.777	m
		Stair section c =	0.000	m
	2	**Effective Span**		
		Considering longest span of the Flight for Design,		
		leff=c/c length	2843.819	mm
	3	**Effective depth d**		
		Using deflection control criteria		
IS		α= 23		
456:2000		γ=δ=β=d=leff/ .5		
			82.42952	
		Assume clear cover =	15	mm
		using bar dia of =	12	mm

	Required depth Dreq =	103.4295	mm			
	Provide overall depth D =	125	mm> Dreq			
	effective depth d =	104	mm			
4	**Load Calculation**					
i)	Load on Landing					
	Self weight = γcx volm=	3.125	KN/m²	3.75		
	Live load=	3	KN/m²	3		
	Floor finish =	1.25	KN/m²	1.28		
	Total load on landing W =	7.375	KN/m²	8.03		
	Factored load Wu = 1.5xW=	11.0625	KN/m²	Wl=		
	Load per unit meter =	**11.06**	**KN/m**	**11.06**	**KN/m**	
ii)	Load on Going					
	Self weight = γc*D*(lin/ln)=	3.815	KN/m²			
	Self weight of step =γc*R/2=	2.223	KN/m²			
	Live Load LL =	3	KN/m²			
	Floor finish FF =	1.25	KN/m²			
	Total load on going W =	10.28705	KN/m²			
	Factored load Wug =	15.43057	KN/m²	Wg=		
	Load per unit meter =	**15.43**	**KN/m**	**15.43**	**KN/m**	

5 Bending moment Calculation



Loadin g on Stair

IS 456:2000 cl.33.2

Fy =Va+Vb=11.06*1.066+15.43*1.777+11.06*0

= 39.22 KN

Taking moment about A,

Vb=(10.49*a²/2+14.89*b*(a+b/2)+10.49*c*(a+b+c/2))/(a+b+c)

= **21.067 KN per meter width**

Va=Fy-Vb= **18.155 KN per meter width**

Let point of zero SF occurs at x distance from A.

= Va -11.06*xa-15.43x(X-a)

X= 1.479 m from A

Since maximum bending moment occurs at zero shear,

BMmax= Va*X-11.06*a²/2-15.43*(X-a)*(a+(X-a)/2) Mumax

Mumax 12.46 KN-m per meter unit width **12.46 KN-m**

For limiting bending moment

Mu*lim=0.138 fck bd²*= 29.85216 KN-m per m unit width

$M_{umax} \leq M_{ulim}$

Here,

so, one-way slab can be designed.

6 Design for Main Reinforcement

Mu=0.87 fy*Ast*d*(1−fyAst/(fck*bd))

29.85*[10]^6=0.87*500*Ast*104*(1−(500*Ast)/(20*104*1000))

OK

Adopting 12mm dia. reinforcement.

Bar dia. 12 mm

Spacing required Sv =

Sv= 1000*ast/Ast

Sv= 155.7321

Providing Spacing of 150mm c/c

Spacing Sv= 150 mm

Hence, **Provide 12mm ϕ bar at 150mm C/C**

Provided Ast=1000*ast/Sv

= 753.98 mm²

Provide 12mm ϕ bar at 150mm C/C

7 Check for deflection control criteria

IS 456:2000 cl.23.2.2.c

α= 23

β= 1

For γ,

$fs=0.58*fy*(Ast\ req)/Astprovided$

Ast %=100Ast/bD= 0.60 %

IS 456:2000 Fig 4

= 279.3258 N/mm²

From fig 4 of IS 456:2000,

γ= 1.6

From fig 5 of IS 456:2000,

δ= 1

λ= $dreq=leff$ 1

77.28 mm<dprovided **Safe**

OK

8 Design for distribution Reinforcement

Ast min. = 0.12%bD= 124.8 mm²

Bar dia. Provided = 8 mm

Spacing Sv=1000*ast/Ast

402.77 mm

Provide 10mmdia bar @ 150mmc/c

hence Sv = 150 mm

Ast provided = 201.0619 mm²

Providmm ϕ bar @150 mm c/c spacing

9 Check For Shear

IS 456:2000

Max. shear force, τumax= 18.155 KN

Nominal Shear stress, τu=Vu/bD

Table 20

Table 19

Provided Ast % pt=(Ast prov.)/(b*d 0.14524 N/mm²

pt %= 0.16085 %

Design Shear stress τc = 0.328 N/mm²>τu

Ok

Maximum Shear stress τvmax = 2.8 N/mm²>τu **Safe**

10 Check for Development Length

Ld=(σs*φ)/(1.6*4*τbd

)=(0.87*500*10)/(1.6*4*1.2)

IS 456:2000 cl.26.2.1.1

Ld= 679.6875 mm

Ld=1.3*M1/V+Lo=1.3*24.14*1000/28.851+Lo

1.3*M1/V= 892.3656 mm>Ld, hence Lo is not required **OK**